

Meromorphic travelling waves for pure-power equations: one pole orbit and a two-wave bound

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1 October 2026

Abstract

The equation $P(d/dz)v + v^n/n = 0$, where P is a polynomial of degree p with complex coefficients and $n \geq 2$, describes the travelling waves of evolution equations with a pure-power nonlinearity and arbitrary linear dispersion and dissipation. We prove that, when $p \geq 3$, every nonconstant solution meromorphic in the whole complex plane has exactly one pole modulo its twisted periods, the translations that multiply it by a root of unity. Hence it is rational, rational in an exponential, or elliptic, and lies in an explicit finite-dimensional space. No condition on the coefficients is imposed; in particular, the equation need not determine the Laurent series at a pole. Up to translation and multiplication by a root of unity, one equation has at most two such solutions, and two do occur. It has at most one unless $n \leq 3$ and P is even. Once the twisted periods of a solution are known, its power v^{n-1} is determined by P ; for elliptic solutions this gives an explicit necessary condition on the coefficients.

Keywords. Meromorphic travelling waves; exact solutions; twisted periods; Laurent series; elliptic functions of the second kind.

MSC2020. 34M05, 35C07, 33E05.

1 Introduction

1.1 The equation and its waves

Exact travelling waves of nonlinear evolution equations are usually found with an ansatz: a polynomial in $\tanh$, a combination of sech and $\tanh$, a Jacobi or Weierstrass elliptic function, or a rational function of an exponential [21, 22, 31]. Every such ansatz produces a function that is meromorphic in the whole complex plane and belongs to the class

$$\mathcal{W} = \{\text{rational functions of } z\} \cup \{\text{rational functions of } e^{kz}, k \neq 0\} \cup \{\text{elliptic functions}\}.$$

Two questions arise. Can a travelling wave be meromorphic in $\mathbb{C}$ without belonging to $\mathcal{W}$? And which functions of $\mathcal{W}$, and how many of them, solve one given equation?

We study both questions for the ordinary differential equation

$$P(D)v + \frac{v^n}{n} = 0, \quad D = \frac{d}{dz}, \quad (1.1)$$

where $n \geq 2$ and P is a monic polynomial of degree p with complex coefficients. For $p \geq 3$ the answer to the first question is no: all nonconstant solutions that are meromorphic in $\mathbb{C}$ belong to $\mathcal{W}$. For odd p this follows from a theorem of Eremenko [14, Theorem 2]; for even p it was known

for $P(D) = D^p + a_0$ [15] and for the even operators of degree four [5, 10]. As to the second, these solutions lie in explicit finite-dimensional spaces, and up to the symmetries of the equation there are at most two of them.

Equation (1.1) is the travelling-wave equation of evolution equations such as

$$U_t + U^{n-1}U_x + \sum_{j=1}^p \alpha_j \partial_x^{j+1} U = 0, \quad \alpha_p \neq 0. \quad (1.2)$$

Indeed, travelling waves $U(x, t) = w(x - st)$ of (1.2) satisfy, after one integration,

$$\sum_{j=1}^p \alpha_j w^{(j)} - sw + \frac{w^n}{n} + C = 0$$

with a constant C . When $C = 0$, the substitution $w = \alpha_p^{1/(n-1)} v$ gives (1.1) with $P(\rho) = \alpha_p^{-1}(\sum_j \alpha_j \rho^j - s)$. So the speed s enters the constant term of P , and P is an even polynomial exactly when (1.2) contains only derivatives of odd order, that is, when the evolution equation has no dissipative terms. For $n = 2$ a shift of w removes the constant C and changes only the constant term of P . For $n \geq 3$ we take $C = 0$; when $C \neq 0$, moving a constant solution to 0 produces lower powers of v , and Appendix E treats such nonlinearities.

For example, the Kuramoto–Sivashinsky [27, 34] and Kawahara [19] equations and the Nikolaevskiy equation with dispersion [23] give $n = 2$ with $p = 3$, $p = 4$ and $p = 5$ respectively, and the Swift–Hohenberg equation [38] gives $n = 3$ with $p = 4$. The last equation, like reaction–diffusion equations, is not of the form (1.2): its travelling-wave equation is used as it stands, without an integration, so p equals the order of the equation, and the speed enters as the coefficient of D (Section 9.3).

Throughout the paper, a *wave* is a nonconstant solution of (1.1) that is meromorphic in the whole complex plane. Suppose a wave v has a pole of order q at z_0 , with leading coefficient c . The highest derivative $v^{(p)}$ must balance the nonlinear term v^n/n , which gives

$$p = (n-1)q, \quad c^{n-1} = -n(-1)^p(q)_p, \quad (1.3)$$

where $(a)_k = a(a+1) \cdots (a+k-1)$ is the rising factorial. A solution without poles is constant (Proposition 4.3). Hence waves exist only when $n-1$ divides p . All poles of all waves then have the same order $q = p/(n-1)$, and the $n-1$ possible leading coefficients differ by factors ω with $\omega^{n-1} = 1$.

Equation (1.1) is invariant under translations and under the *phase changes* $v \mapsto \omega v$, where ω lies in the group μ_{n-1} of $(n-1)$ st roots of unity. Combining the two gives the notion around which this paper is organized.

Definition 1.1. A *twisted period* of a wave v is a number $\gamma \in \mathbb{C}$ such that

$$v(z + \gamma) = \chi(\gamma) v(z) \quad \text{for some } \chi(\gamma) \in \mu_{n-1}.$$

The number $\chi(\gamma)$ is the *multiplier* of v under the translation $z \mapsto z + \gamma$. The twisted periods form a group Γ , and $\chi: \Gamma \rightarrow \mu_{n-1}$ is a homomorphism, the *character* of v . Its kernel is the group of ordinary periods.

For instance, $\operatorname{sech} z$ changes sign under $z \mapsto z + i\pi$, whereas $\tanh z$ is unchanged: $i\pi$ is a twisted period of sech with multiplier -1 , and an ordinary period of $\tanh$. Both functions have a single pole modulo $i\pi$; modulo its ordinary period $2\pi i$, $\operatorname{sech} z$ has two poles, with opposite residues. Twisted periods that are not ordinary periods can occur only for $n \geq 3$, since $\mu_1 = \{1\}$.

1.2 Results

In the statements we write $m = n - 1$ and $q = p/m$.

Theorem A (One pole orbit). *Let $n \geq 2$ and let P have degree $p \geq 3$. Every wave v of (1.1) has exactly one pole modulo its twisted periods: if z_0 is a pole, then the set of all poles is the orbit $z_0 + \Gamma$. After the translation that moves z_0 to 0, the wave has one of the following forms.*

- (i) $\Gamma = \{0\}$, and v is a polynomial of degree q in $1/z$ without constant term.
- (ii) $\Gamma = (2\pi i/k)\mathbb{Z}$ for some $k \neq 0$. If $e^{2\pi i r/m}$, with $1 \leq r \leq m$, is the multiplier of v under $z \mapsto z + 2\pi i/k$, then

$$v = \beta + \sum_{j=0}^{q-1} d_j D^j g_r, \quad g_r(z) = \frac{k e^{rkz/m}}{e^{kz} - 1}, \quad (1.4)$$

with constants $\beta, d_0, \dots, d_{q-1}$, where $d_{q-1} \neq 0$, and $\beta = 0$ unless $r = m$.

- (iii) Γ is a lattice, and v is an elliptic function of the second kind: $v(z + \gamma) = \chi(\gamma)v(z)$ for $\gamma \in \Gamma$, with poles of order q at the points of Γ and nowhere else.

In particular, every wave belongs to $\mathcal{W}$.

The last assertion holds because the ordinary periods form a subgroup of index at most m in Γ : the character maps Γ into μ_m . So a wave has at most $m = n - 1$ poles modulo its ordinary periods; in case (ii) it is a rational function of $e^{kz/m}$, and in case (iii) it is elliptic for a sublattice of Γ . By the forms (i)–(iii), the search for all waves of a given equation reduces to substitutions with finitely many unknown coefficients; Section 9.3 carries one out.

In (1.4), the function g_r has simple poles exactly at the points of $(2\pi i/k)\mathbb{Z}$, with residue 1 at 0; it is multiplied by $e^{2\pi i r/m}$ under $z \mapsto z + 2\pi i/k$, and it stays bounded as $\text{Re}(kz) \rightarrow \pm\infty$. Its logarithmic derivative is $rk/m - G$, where

$$G = g_m = \frac{k e^{kz}}{e^{kz} - 1} = \frac{k}{2} \left(1 + \coth \frac{kz}{2} \right), \quad DG = G(k - G).$$

By induction, $D^j g_r$ is g_r times a polynomial of degree j in G ; so (1.4) says that $v = \beta + g_r Q(G)$ for a polynomial Q of degree $q - 1$. For $n = 2$ this is a polynomial of degree q in G , and the translation $z \mapsto z + i\pi/k$ turns it into a polynomial in $\tanh(kz/2)$, the form assumed by the tanh method. For $n = 3$ and $r = 1$, the same translation turns g_1 into a multiple of $\text{sech}(kz/2)$, and v into sech times a polynomial in $\tanh$. In case (iii) the space of such elliptic functions has an explicit basis (Lemma 5.3).

In case (ii) we call $\text{Re}(kz) \rightarrow \pm\infty$ the two *ends* of the wave (for real k these are $z \rightarrow \pm\infty$ along the real axis) and k its *rate*; replacing k by $-k$ interchanges the two ends. A wave is *twisted* if its character is nontrivial, that is, if some twisted period is not an ordinary period; otherwise it is *untwisted*.

Theorem B (At most two waves). *Let $n \geq 2$ and let P have degree $p \geq 3$. Up to translation and phase change, equation (1.1) has at most two waves. It has at most one unless $n \leq 3$ and P is even. For $P(D) = D^4 - s$ with $s \neq 0$ it has exactly two, both for $n = 2$ and for $n = 3$.*

The theorem counts all waves, complex and unbounded ones included. It bounds their number; which operators admit a wave is a different question (Section 10). In terms of (1.2): at a fixed

speed and a fixed integration constant, an evolution equation of order $p + 1 \geq 4$ has at most two meromorphic travelling waves up to translation and phase change, and at most one if it contains a dissipative term or if $n \geq 4$. Here the constant is zero when $n \geq 3$. For $n = 2$ it is arbitrary, but it has to be fixed, because P depends on it (Section 8.2). Section 2 exhibits the two waves of $D^4 - s$ for $n = 3$; one of them is twisted.

The third result shows how much of a wave the operator determines before the wave is known. We normalize a wave so that it has a pole at 0 with leading coefficient c , where c is a fixed root of (1.3), and put $u = v/c = z^{-q} + O(z^{1-q})$.

Theorem C (The power of a wave). *Let $p \geq 3$. The coefficients of $z^{-p}, \dots, z^{-1}, z^0$ in the Laurent series of u^{n-1} at 0 are polynomials $\ell_0 = 1, \ell_1, \dots, \ell_p$ in the coefficients of P ; they are the same for all waves. Together with the group Γ they determine u^{n-1} by the explicit formula of Theorem 8.1. In particular, the residue $\mathcal{R}(P) = \ell_{p-1}$ of u^{n-1} at 0 satisfies*

$$\mathcal{R}(P) = 0 \quad \text{if } \Gamma \text{ is a lattice,} \quad u_+^{n-1} - u_-^{n-1} = k \mathcal{R}(P) \quad \text{if } \Gamma = (2\pi i/k)\mathbb{Z},$$

where $u_{\pm}$ are the limits of u at the ends $\operatorname{Re}(kz) \rightarrow \pm\infty$.

For example, the Kuramoto–Sivashinsky equation with dispersion, $U_t + UU_x + \mu U_{xx} + \delta U_{xxx} + \nu U_{xxxx} = 0$, has $\mathcal{R} = (16\mu\nu - \delta^2)/(608\nu^2)$; so its elliptic travelling waves require $\delta^2 = 16\mu\nu$ (Section 8.2).

Three further results follow from the theorems (Sections 8 and 9). If P is even, every wave is even or odd about each of its poles, and a *front*, that is, a wave with different limits at its two ends, exists only for odd q . If $n \geq 3$ and P is an even polynomial of degree $p \geq 4$ plus a nonzero multiple of D , there are no waves at all. And two waves of one equation that are not related by translation and phase change never have the same twisted periods.

The theorems need $p \geq 3$. For $p = 2$ one equation can have infinitely many waves, and some equations have waves outside $\mathcal{W}$. The one-pole conclusion still holds for $p = 1$, for even P of degree two, and for all elliptic waves with $p = 2$ (Appendix D).

The proofs. The proofs use Laurent series, Cauchy's estimate, and the identity theorem. At a pole, the Laurent series of a wave is determined by P up to a root of unity and, when p is even, one number h , the *free coefficient*. If the series at two poles agree up to a root of unity, the difference of the poles is a twisted period. So a wave whose poles all carry the same h has one pole modulo Γ , and two waves that share a value of h coincide after a translation and a phase change. To find the values of h we form combinations of a wave and its derivatives that have no poles. By Cauchy's estimate a wave is large only near its poles; so such a combination is constant as soon as it takes finitely many values at the poles. Its Taylor coefficients of positive order at a pole then vanish, and they are polynomials in h whose coefficients depend only on P . This gives an equation for h of degree one when $n \geq 4$ and of degree two when $n \leq 3$, and a further equation of degree one when P is not even. A count of zeros and poles shows that one wave cannot carry both roots of the quadratic equation. Section 2 carries out the argument on one equation, and Appendix A lists the notation.

1.3 Earlier work

Odd degree. For odd p there is no free coefficient, and Eremenko [14, Theorem 2] showed by Nevanlinna theory [7, 28] that all meromorphic solutions belong to $\mathcal{W}$, with at most $n - 1$ poles per period; the argument that two poles with equal Laurent series differ by a period is his [13, 14].

For $n = 2$ the forms (i)–(iii) are those of Demina and Kudryashov [11, Theorem 1], who also note that an equation with one Laurent series has at most one meromorphic solution up to translation. The case $n = 4$, $p = 3$, with lower powers in the nonlinearity, is due to Conte and Ng [4].

Even degree. The free coefficient was noted by Eremenko [13]; Lemma 3.1 is Proposition 2.1 of Wu [39], which covers every equation with the dominant terms $v^{(p)}$ and v^n . For $P(D) = D^p + a_0$ (in their setting, D^p with a general nonlinearity), Eremenko, Liao, and Ng [15, Lemma 4] fixed this coefficient by the first integral that comes from the integrating factor v' . In the same way all meromorphic solutions were found for even operators of degree four: for $n = 2, 3, 5$ by Demina and Kudryashov [10], for $n = 3$ with a general cubic nonlinearity by Conte, Ng, and Wong [5, Theorem 1], and for $n = 5$ with a cubic term by Wu [39, Theorem 2.1]. Their lists contain the waves that we call twisted, and the two-wave examples of Sections 2 and 8.3. The same argument works for every even operator (Appendix E); for an operator that is not even this integral is not available. For $n = 2$ and operators of degree four that are not even, Demina [9] classified the solutions that are elliptic or rational in an exponential, and each of them has one pole per period [3, Section 5.6, Theorem 5.8].

Examples and tools. The Swift–Hohenberg waves of Section 9.3 were found by Kudryashov and Sinelshchikov [25], for an equation with an additional quadratic term; they also show that without dispersion all meromorphic travelling waves are stationary. The elliptic waves of the Kuramoto–Sivashinsky equation are due to Kudryashov [20], and the condition $\delta^2 = 16\mu\nu$ to Eremenko [14]; it is an instance of the residue criterion for elliptic solutions [4, 18], which Theorem C applies to u^{n-1} . For $p = 2$, waves outside $\mathcal{W}$ were found by Ablowitz and Zeppetella [1]; see also [6, Lemmas 4.4 and 4.5]. The estimates of Section 4 rest on the weighted maximum of Zalcman’s rescaling lemma [40, 41], which Zalcman traces back to Landau’s proof of Bloch’s theorem; Lemma 4.2 is a pointwise counterpart of Clunie’s lemma [28], and Poláčik, Quittner, and Souplet [33] prove estimates of the same form for superlinear elliptic equations. For the Painlevé transcendents, Steinmetz [35–37] uses estimates away from the poles and combinations of a solution and its derivative that are regular at the poles; expressions without poles built from the first Laurent coefficients are used, with Nevanlinna theory, by Halburd and Wang [16] and Meng and Halburd [29]. Functions of the second kind and Hermite’s decomposition are classical [2, 11, 17].

New here. Theorem A for operators of even degree that are not even, for every n and for an arbitrary meromorphic solution; for $n \geq 3$, the single pole modulo twisted periods; the bound of Theorem B beyond odd p and even operators of degree four, with the fact that two waves require an even operator and $n \leq 3$; the formula of Theorem C for u^{n-1} , with the relation $u_+^{n-1} - u_-^{n-1} = k\mathcal{R}(P)$ between the limits at the two ends; and the completeness of the list of Section 9.3. The proofs use Cauchy’s estimate in place of Nevanlinna theory.

2 The argument on one equation

We first carry out the whole argument on the equation $v'''' - sv + v^3/3 = 0$, that is, $n = 3$ and $P(D) = D^4 - s$; it is the travelling-wave equation of $U_t + U^2U_x + U_{xxxx} = 0$ at speed s with zero integration constant. There are seven steps. Steps 1–5 are carried out for every operator in Sections 3–7, and Step 7 in Section 8; each of these steps names the statements that replace it. Step 6 exhibits the two waves of this equation.

2.1 Step 1: the Laurent series at a pole

By (1.3), every pole of a wave is double and its leading coefficient c satisfies $c^2 = -360$. For either root c , the function $u = v/c$ satisfies

$$u'''' = s u + 120 u^3. \quad (2.1)$$

Put a pole at 0 and write $u = z^{-2}(1 + b_1 z + b_2 z^2 + \dots)$. For every $j \geq 1$, the coefficients of z^{j-6} on both sides give

$$((j-2)(j-3)(j-4)(j-5) - 360) b_j = \text{a polynomial in } s \text{ and } b_1, \dots, b_{j-1}.$$

The product $(j-2)(j-3)(j-4)(j-5)$ is 24 or 0 for $j \leq 5$ and increases for $j \geq 6$; it equals $360 = 3 \cdot 4 \cdot 5 \cdot 6$ only for $j = 8$. So the factor on the left vanishes only there, and $b_1, \dots, b_7$ are determined by s . At $j = 8$ the coefficient b_8 drops out, and the other terms, $-b_4(s + 360b_4)$, cancel because $b_4 = -s/360$. So b_8 can be any number h , and the later coefficients are determined by s and h . The result is the series

$$f(z; h) = z^{-2} - \frac{s}{360} z^2 + h z^6 + O(z^{10}),$$

and at every pole z_i of every wave

$$v(z_i + z) = \pm c f(z; h_i) \quad \text{for some number } h_i. \quad (2.2)$$

We call h_i the *free coefficient* of the wave at z_i . It is all that the equation leaves open near a pole, and it enters the series late, at z^6 . The rest of the argument is about its values.

In general (Section 3): for odd p there is no free coefficient; for even p there is one, at z^{nq} (Lemma 3.1, Corollary 3.3).

2.2 Step 2: a wave is large only near its poles

We quote one estimate, proved in Section 4: every wave satisfies $|v(z)| \leq C(1 + d(z)^{-2})$, where $d(z)$ is the distance from z to the nearest pole and C depends only on s (Lemma 4.2). The reason is that on a disk without poles where $|v|$ is large, Cauchy's estimate makes v'''' too small to balance $v^3/3$. Two consequences are used.

- (a) If the poles of v form finitely many cosets $z_i + \Gamma$ of its group of twisted periods, then v is elliptic, or a rational function of an exponential that is bounded at both ends, or a rational function bounded at infinity, according as Γ is a lattice, a group $\tau\mathbb{Z}$ with $\tau \neq 0$ (rank one), or $\{0\}$ (Lemma 5.1(c)).
- (b) A polynomial in $v, v', v'', \dots$ that has no poles and is bounded on the set of poles of v is constant (Lemma 4.4).

2.3 Step 3: equal series give a twisted period

If two poles z_1, z_2 of a wave have the same free coefficient, then $v(z_2 + z) = \pm v(z_1 + z)$ near $z = 0$ by (2.2), hence for all z by the identity theorem: $z_2 - z_1$ is a twisted period. For ordinary periods this argument is due to Eremenko [13, 14]. So the poles of a wave that carry one value of h form a single coset $z_1 + \Gamma$. In the same way, two waves that have poles with the same value of h differ by a translation and a sign.

Hence a wave whose poles all carry the same value of h has one pole modulo Γ , and by Step 2(a) it is elliptic, rational in an exponential, or rational. It remains to find the possible values of h (Step 4) and to show that one wave carries only one of them (Step 5).

In general (Section 5): Corollary 5.2 for equal series, and Proposition 5.4 for the forms (i)–(iii) of Theorem A. For odd p there is no free coefficient, and this step already proves Theorems A and B (Section 5.3).

2.4 Step 4: an expression without poles

The values of h are found from a polynomial in u and its derivatives that has no poles. The combination $uu'' - \frac{3}{2}(u')^2$ vanishes for $u = z^{-2}$. On the series f it therefore starts with the next coefficient: $ff'' - \frac{3}{2}(f')^2 = -\frac{s}{18}z^{-2} + O(z^2)$. We want to cancel this pole by a multiple of a derivative of u , at every pole of u and for either sign in (2.2). A derivative of u changes sign with u , and the quadratic combination does not. So we first multiply by f , which makes the expression odd in u , and then compare with $f'' = 6z^{-4} + O(1)$:

$$f^2 f'' - \frac{3}{2} f (f')^2 = -\frac{s}{18} z^{-4} + O(1), \quad \frac{s}{108} f'' = \frac{s}{18} z^{-4} + O(1).$$

Neither principal part contains h , and they cancel in the sum. So

$$\Psi = u^2 u'' - \frac{3}{2} u (u')^2 + \frac{s}{108} u''$$

has no pole at a pole of u , whichever sign occurs in (2.2): it is an entire function. With more terms of f one finds, at a pole z_0 where $u(z_0 + z) = \pm f(z; h)$,

$$\Psi(z_0 + z) = \pm \left[72h + \frac{7s^2}{97200} + \frac{7(388800h - s^2)(6998400h + 7s^2)}{365316480000} z^8 + O(z^{12}) \right].$$

The value of Ψ at a pole contains the unknown h , but Ψ' vanishes at every pole of u . By Step 2(b), Ψ' is constant, hence zero, and Ψ is constant. The coefficient of z^8 must therefore vanish:

$$h = \frac{s^2}{388800} \quad \text{or} \quad h = -\frac{7s^2}{6998400}.$$

Every pole of every wave carries one of these two values of the free coefficient.

In general (Section 6): two expressions in u whose principal parts do not contain h are combined so that their poles cancel. This gives an equation for h of degree one if $n \geq 4$ (Proposition 6.4) and of degree two if $n = 2, 3$ (Proposition 6.6).

2.5 Step 5: one wave cannot carry both values

Suppose a wave v has poles z_1 and $z_2 = z_1 + a$ with free coefficients $h_1 \neq h_2$. By Step 4 every pole of v carries h_1 or h_2 , so by Step 3 its poles form the two cosets $z_1 + \Gamma$ and $z_2 + \Gamma$, and Step 2(a) applies. Choose the sign in $W(x) = v(z_1 + a + x) \mp v(z_1 + x)$ so that the leading terms at $x = 0$ cancel. The two series (2.2) agree up to the term hz^6 , so $W(x) = \pm c(h_2 - h_1)x^6 + \dots$. Thus W has a zero of order 6 at 0, and at every point of Γ , because W has the same multipliers as v . Its poles lie in the cosets $\pm a + \Gamma$ and are at most double. If Γ is a lattice, then W^2 is a function on the torus $\mathbb{C}/\Gamma$ with a zero of order 12 and poles of total order at most $4 + 4$. This is impossible: a nonzero meromorphic function on a compact Riemann surface has as many zeros as poles. If Γ has rank one or is $\{0\}$, then v is bounded at the two ends of the cylinder $\mathbb{C}/\Gamma$, or at infinity,

by Step 2(a); so W^2 extends to a sphere, and the same count applies. For an even operator such as $D^4 - s$, the conclusion also follows from the first integral obtained with the factor v' : it takes one value on the wave, and this value is affine in h with a nonzero coefficient (Lemma E.1 and Theorem E.2; for $D^p + a_0$ see Eremenko, Liao, and Ng [15, Lemma 4]). The count of zeros and poles works for every operator.

So all poles of a wave carry the same value of h . By Step 3 the wave has one pole modulo Γ and is elliptic, rational in an exponential, or rational: this is the first assertion of Theorem A for (2.1), and the forms (i)–(iii) follow as in Proposition 5.4. Each of the two values of h belongs to at most one wave up to translation and sign: this is the bound of Theorem B.

In general (Section 7): Lemma 7.1 for two values in one wave, where $n = 2$ needs the odd part of W . If $n = 2, 3$ and P is not even, a further equation of degree one leaves a single value of h (Proposition 7.3).

2.6 Step 6: the two waves

Both values of h occur. For $s = 108$ equation (2.1) is

$$u'''' = 108u + 120u^3, \quad (2.3)$$

and two solutions are

$$u_1 = \wp(z; -6, 0), \quad u_2 = -y', \quad y^2 = \wp(z; 4, 0), \quad y = z^{-1} + O(z^3). \quad (2.4)$$

Here $\wp(z; g_2, g_3)$ is the Weierstrass function with $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$; both discriminants $g_2^3 - 27g_3^2$ are nonzero, so these are elliptic functions and not degenerate ones. The Weierstrass identities give $u_1'' = 6u_1^2 + 3$ and then $u_1'''' = 120u_1^3 + 108u_1$. The function $\wp(z; 4, 0)$ has double poles at the points of its period lattice Γ_0 and double zeros at the half-periods where $\wp = 0$, so it has a meromorphic square root y on $\mathbb{C}$. From $(y')^2 = y^4 - 1$ and $y'' = 2y^3$ we get $u_2^2 = y^4 - 1$ and $u_2'''' = (120y^4 - 12)u_2 = 120u_2^3 + 108u_2$.

The solution u_1 has trivial character; u_2 is twisted. Indeed, y has a single simple pole in a period cell of Γ_0 , so it is not periodic for Γ_0 : its multipliers under Γ_0 form a character χ of order two, and y is an ordinary elliptic function for the sublattice $\ker \chi$ of index two, with two simple poles in a period cell. The function $u_2 = -y'$ has the same character: its twisted periods are $\Gamma = \Gamma_0$ and its ordinary periods are $\ker \chi$, as in Theorem A(iii).

The Laurent expansions are

$$\begin{aligned} u_1 &= z^{-2} - \frac{3}{10}z^2 + \frac{3}{100}z^6 + O(z^{10}), \\ u_2 &= z^{-2} - \frac{3}{10}z^2 - \frac{7}{600}z^6 + O(z^{10}), \end{aligned} \quad (2.5)$$

so the free coefficients are $h_1 = 3/100$ and $h_2 = -7/600$, the two values of Step 4 for $s = 108$. So $v_1 = cu_1$ and $v_2 = cu_2$, with $c^2 = -360$, are two waves that are not related by translation or phase change. They are all the waves of this equation: a wave has a pole, by Step 4 its free coefficient there is h_1 or h_2 , and by Step 3 the wave is then v_1 or v_2 up to a translation and a sign.

If u solves (2.1), then $\lambda^2 u(\lambda z)$ solves it with $\lambda^4 s$ in place of s . So $c\lambda^2 u_{1,2}(\lambda z)$ are the two waves for $P(D) = D^4 - 108\lambda^4$. Every $s \neq 0$ has the form $108\lambda^4$; hence $v'''' - sv + v^3/3 = 0$ has exactly two waves for every $s \neq 0$, as Theorem B asserts for $n = 3$ (Section 8.3 treats $n = 2$). For $s = 0$ the two values of Step 4 coincide, $h = 0$; the series $f(z; 0)$ is then z^{-2} , and the only wave is cz^{-2} . The two waves are the solutions (36) and (51) of Demina and Kudryashov [10] with $\beta = 0$ and $C_1 = 0$, and the solutions (17) and (22) of Conte, Ng, and Wong [5] with $b = s_1 = s_3 = 0$, in the notation of those papers.

2.7 Step 7: the square of a wave

The series of Step 1 also determines the square of a wave, once Γ is known. It gives $u^2 = z^{-4} - \frac{s}{180} + O(z^4)$ at a pole of any wave: in the notation of Theorem C, $\ell_0 = 1$, $\ell_1 = \ell_2 = \ell_3 = 0$ and $\ell_4 = -s/180$. The function u^2 is invariant under Γ , because the multipliers of u are ± 1 , and it has one pole modulo Γ . Let Γ be a lattice and $\wp = \wp(z; g_2, g_3)$ its Weierstrass function, so that $\wp^2 = z^{-4} + g_2/10 + O(z^2)$. Then $u^2 - \wp^2$ is an elliptic function without poles, hence a constant:

$$u^2 = \wp^2 - \frac{g_2}{10} - \frac{s}{180}.$$

So the square of a wave is known as soon as its lattice is known; the character does not enter. For $s = 108$ the formula gives $u_1^2 = \wp(z; -6, 0)^2$ and $u_2^2 = \wp(z; 4, 0)^2 - 1$, in agreement with Step 6.

In general (Section 8): Theorem 8.1 gives u^{n-1} for each kind of Γ , and Corollary 8.2 the conditions on its residue, which is $\ell_3 = 0$ here.

3 The Laurent series at a pole

This section does Step 1 of Section 2 for every operator. From now on $p/(n-1)$ is a positive integer, as it must be if a wave exists (Proposition 4.3). We fix the notation

$$m = n - 1, \quad p = mq, \quad K = (-1)^p (q)_p = (-1)^p \frac{(nq - 1)!}{(q - 1)!},$$

and a number c with $c^m = -nK$, as in (1.3). If a wave v has a pole at z_0 , a phase change makes its leading coefficient there equal to c , and then $u(z) = v(z_0 + z)/c$ satisfies

$$P(D)u = K u^n, \quad u = z^{-q}(1 + O(z)). \quad (3.1)$$

We call such a u a *normalized wave*; it has its pole at 0.

We write

$$P(\rho) = \rho^p + a_{p-1}\rho^{p-1} + \cdots + a_1\rho + a_0.$$

If u is a normalized wave for P and $\lambda \neq 0$, then $\lambda^q u(\lambda z)$ is a normalized wave for $P_\lambda(\rho) = \lambda^p P(\rho/\lambda)$. This *scaling* multiplies a_{p-k} by λ^k ; accordingly we say that a_{p-k} has *weight* k .

Let u be a normalized wave and write $u = z^{-q} \sum_{j \geq 0} b_j z^j$ with $b_0 = 1$. The coefficient of z^{j-q-p} in $P(D)u - K u^n$ gives the recurrence

$$I(j) b_j = F_j(b_1, \dots, b_{j-1}), \quad I(j) = (j - q)(j - q - 1) \cdots (j - q - p + 1) - nK. \quad (3.2)$$

The left-hand side collects the terms that are linear in b_j , coming from D^p and from u^n . The right-hand side F_j collects the rest; it is a polynomial in $b_1, \dots, b_{j-1}$ and in the coefficients of P . The same recurrence holds at every pole of every wave, because (3.1) does not depend on which root c was chosen. Which coefficients it determines depends on the integer roots of the *indicial polynomial* I .

Lemma 3.1 (The resonance). *If p is odd, then $I(j) \neq 0$ for every integer $j \geq 1$. If p is even, the only positive integer root of I is*

$$J = p + 2q = (n + 1)q.$$

Proof. Write $\Pi(j) = (j - q)(j - q - 1) \cdots (j - q - p + 1)$ for the product in $I(j)$. Since $q + p = nq$, we have $n(q)_p = (q + 1)_p$. For $1 \leq j < q$, all factors of $\Pi(j)$ are negative, and $|\Pi(j)| = (q - j)_p < (q)_p < n(q)_p = |nK|$, so $I(j) \neq 0$. For $q \leq j < q + p$, one factor vanishes and $I(j) = -nK \neq 0$. For $j \geq q + p$, we have $\Pi(j) = (j - q - p + 1)_p > 0$, and this is strictly increasing in j . If p is odd, then $-nK = n(q)_p > 0$, so $I(j) > 0$. If p is even, then $nK = (q + 1)_p$, and $I(j) = 0$ exactly when $j - q - p + 1 = q + 1$, that is, $j = p + 2q$. $\square$

In the language of the Painlevé test, the roots of I are the resonances; for even p the only positive integer resonance is J [39]. So for odd p the recurrence determines all b_j , and the Laurent series of a normalized wave is determined by P . For even p the recurrence determines $b_1, \dots, b_{J-1}$; at the index J it reads $F_J(b_1, \dots, b_{J-1}) = 0$ and does not contain b_J ; and it determines b_j for $j > J$ once b_J is given. For every operator P of even degree and every $h \in \mathbb{C}$ we define the series

$$f(z; h) = z^{-q} \sum_{j \geq 0} b_j(h) z^j, \quad b_0 = 1, \quad b_J = h, \quad (3.3)$$

by the recurrence at all indices $j \neq J$. Then $P(D)f - Kf^n = -F_J z^q$, where the number F_J depends on P and not on h . If P admits a wave, the wave has a pole (Proposition 4.3), and its Laurent series there satisfies the recurrence at the index J as well. So $F_J = 0$, and $f(z; h)$ is a formal solution of (3.1) for every h . (We define the series for every operator, because Sections 6 and 7 treat the quantities built from it as polynomials in the coefficients of P .) At each pole z_i of a wave v we then have

$$v(z_i + z) = c \omega_i f(z; h_i) \quad \text{for some } \omega_i \in \mu_m, \quad h_i \in \mathbb{C}. \quad (3.4)$$

We call h_i the *free coefficient* of the wave at the pole z_i , ω_i its *phase* there, and $f(\cdot; h_i)$ its *normalized series* there. For odd p we use the same notation without h .

Lemma 3.2 (Weights). *Let p be even. Each $b_j(h)$ is a polynomial in h and in the coefficients of P . It is weighted-homogeneous of weight j if each a_{p-i} is given its weight i and h the weight J . In particular, $b_j(h)$ has degree at most $\lfloor j/J \rfloor$ in h ; the part of $f(z; h)$ of degree $k \geq 1$ in h is $O(z^{kJ-q})$; and the part of degree one is $h z^{J-q}(1 + O(z))$.*

Proof. The recurrence gives b_j as a polynomial, because $I(j)$ is a nonzero number for $j \neq J$. For $\lambda \neq 0$, the series $\lambda^q f(\lambda z; h) = z^{-q} \sum_j \lambda^j b_j(h) z^j$ starts with z^{-q} , has the coefficient $\lambda^J h$ at the index J , and satisfies

$$P_\lambda(D)[\lambda^q f(\lambda z; h)] - K[\lambda^q f(\lambda z; h)]^n = \lambda^{nq} (P(D)f - Kf^n)(\lambda z; h) = -\lambda^J F_J z^q.$$

So it satisfies the recurrence of P_λ at all indices $j \neq J$, and it is the series (3.3) of the operator P_λ with free coefficient $\lambda^J h$. The coefficients of P_λ are $\lambda^i a_{p-i}$, which proves the homogeneity. All weights are positive, so a monomial of b_j that contains h^k has $kJ \leq j$. The last two assertions follow, because b_j does not depend on h for $j < J$ and $b_J = h$. $\square$

Corollary 3.3. (a) *The principal part and the constant term of a normalized series are determined by P .* (b) *If z_1, z_2 are poles of a wave v , written as in (3.4), then*

$$v(z_2 + z) - \frac{\omega_2}{\omega_1} v(z_1 + z) = O(z^{nq}).$$

Proof. (a) These are the coefficients b_j with $j \leq q$, and $q < J$ when p is even. (b) The two normalized series have the same coefficients $b_0, \dots, b_{J-1}$, so the difference is $c \omega_2 (f(z; h_2) - f(z; h_1)) = O(z^{J-q})$, and $J - q = nq$. $\square$

Thus the free coefficient appears only at the order z^{nq} , long after the pole; Sections 6 and 7 rest on this.

4 A wave is large only near its poles

This section proves the estimate of Step 2 of Section 2 and its consequence (b): Lemma 4.2 bounds a solution in terms of the distance to its poles, and Lemma 4.4 says that an expression without poles is constant if it is bounded on the poles. The first is used in Section 5 to place a wave in $\mathcal{W}$; the second, in Section 6, to obtain the equations for the free coefficient. They are the only analytic estimates in the paper, and they do not use the form of the nonlinearity. We prove them for the solutions, meromorphic in $\mathbb{C}$, of

$$P(D)v + B(v) = 0, \quad (4.1)$$

where $P(D) = \sum_{j=0}^p a_j D^j$ is any operator with constant coefficients, $a_p \neq 0$ and $p \geq 1$, and B is any polynomial of degree $n \geq 2$; here $q = p/(n-1)$ need not be an integer. This generality is used in Proposition 4.3 and in Appendix E; for Theorems A–C the reader may take $B(v) = v^n/n$. For a solution v , let S be its set of poles and $d(z) = \text{dist}(z, S)$; if S is empty, we put $d \equiv \infty$ and $d^{-q} = 0$.

Both lemmas rest on the maximum of a weighted modulus, the device of Zalcman's rescaling lemma [40, 41]. In a given disk it produces a point z_* at which the modulus of a holomorphic function w , weighted by a power of the distance to the boundary, is at least as large as at the centre, and such that w stays close to $w(z_*)$ on a disk about z_* whose radius is proportional to the distance from z_* to the boundary.

Lemma 4.1 (A weighted maximum). *Let w be holomorphic on a neighbourhood of the closed disk $\overline{D}(z_0, R)$ and not identically zero on it, and let $\theta > 0$. There is a point z_* in the open disk such that, with $\delta = R - |z_* - z_0|$,*

$$R^\theta |w(z_0)| \leq \delta^\theta |w(z_*)| \quad \text{and} \quad \max_{|z - z_*| \leq \delta/2} |w(z)| \leq 2^\theta |w(z_*)|.$$

Moreover, with $c_\theta = 2^{-\theta-3}$,

$$|w(z) - w(z_*)| \leq \frac{1}{3} |w(z_*)| \quad \text{for } |z - z_*| \leq c_\theta \delta.$$

In particular, w has no zero in $\overline{D}(z_*, c_\theta \delta)$.

Proof. Let z_* be a point where $(R - |z - z_0|)^\theta |w(z)|$ attains its maximum on the closed disk. The maximum is positive and the weight vanishes on the boundary, so z_* is an interior point, and the first inequality is the comparison with the value at z_0 . On $\overline{D}(z_*, \delta/2)$ the weight is at least $(\delta/2)^\theta$, which gives the second inequality. Put $M = |w(z_*)| > 0$. Cauchy's estimate on $\overline{D}(z_*, \delta/2)$ bounds the Taylor coefficients of w at z_* , and gives, for $|t| < \delta/2$,

$$|w(z_* + t) - w(z_*)| \leq 2^\theta M \sum_{j \geq 1} \left(\frac{2|t|}{\delta} \right)^j.$$

For $|t| \leq c_\theta \delta$ the right-hand side is at most $M/3$. □

Lemma 4.2 (Distance to the poles). *There are constants C_j , depending only on P , B and j , such that every meromorphic solution of (4.1) satisfies, for $z \notin S$,*

$$|v(z)| \leq C_0(1 + d(z)^{-q}), \quad |v^{(j)}(z)| \leq C_j(1 + d(z)^{-q-j}) \quad (j \geq 1). \quad (4.2)$$

If S is finite, then v is a rational function bounded at infinity.

Proof. For $v \equiv 0$ there is nothing to prove. Choose $C_B > 0$ and $V_0 \geq 1$ such that $|B(x)| \geq C_B|x|^n$ for $|x| \geq V_0$. Fix $z_0 \notin S$ and $0 < R < d(z_0)$. Apply Lemma 4.1 to v on $\overline{D}(z_0, R)$ with $\theta = q$, and put $V = |v(z_*)|$ and $r = \delta/2$. Suppose $V \geq V_0$. Since $|v| \leq 2^q V$ on $\overline{D}(z_*, r)$, Cauchy's estimate gives $|v^{(j)}(z_*)| \leq j! 2^q V r^{-j}$, and the equation gives

$$C_B V^n \leq |B(v(z_*))| = |P(D)v(z_*)| \leq 2^q V \sum_{j=0}^p |a_j| j! r^{-j}.$$

Hence $V^{n-1} \leq C' \max(1, r^{-p})$, and so $V \leq C(1 + r^{-q})$, with C depending only on P and B . If $V < V_0$, the same bound holds as soon as $C \geq V_0$. The first inequality of Lemma 4.1 now gives

$$R^q |v(z_0)| \leq \delta^q V \leq C(\delta^q + 2^q) \leq C(R^q + 2^q).$$

Letting $R \rightarrow d(z_0)$, or $R \rightarrow \infty$ if S is empty, proves the first estimate.

For the derivatives at z , use the disk of radius $r_1 = \frac{1}{2} \min(1, d(z))$. On this disk the distance to S is at least r_1 , so the first estimate bounds v there by $C_0(1 + r_1^{-q})$, and Cauchy's estimate bounds $v^{(j)}(z)$ by $j! r_1^{-j} C_0(1 + r_1^{-q})$. This is the second estimate.

If S is finite, then $d(z) \rightarrow \infty$ as $z \rightarrow \infty$, so v is bounded near infinity. It extends holomorphically to ∞ , and a meromorphic function on the Riemann sphere is rational. $\square$

Proposition 4.3 (Solutions without poles). *Every entire solution of (4.1) is constant. If $p/(n-1)$ is not a positive integer, every meromorphic solution of (4.1) is constant.*

Proof. An entire solution is bounded by Lemma 4.2, hence constant by Liouville's theorem. At a pole of any solution the balance of $v^{(p)}$ with the term of degree n in $B(v)$ gives the order $p/(n-1)$. If this is not a positive integer there is no pole, and the solution is entire. $\square$

For example, the travelling waves of the generalized Bretherton equation $U_{tt} + \alpha U_{xx} + \beta U_{xxxx} + \delta U + \gamma U^n = 0$ with $\beta\gamma \neq 0$ satisfy, after a rescaling, (1.1) with an even operator of degree four, so waves exist only for $n \in \{2, 3, 5\}$ (for elliptic waves and nonzero coefficients of D^2 and D^0 in P , see [26]).

Lemma 4.4 (Bounded on the poles). *Let v be a nonconstant meromorphic solution of (4.1), and let Φ be a polynomial in $v, v', v'', \dots$ with constant coefficients. If Φ has no poles and is bounded on S , then Φ is constant.*

Proof. We may assume $\Phi \not\equiv 0$. Put $M = \sup_S |\Phi|$. By (4.2) there are constants $C, N > 0$ such that $|\Phi(z)| \leq C(1 + d(z)^{-N})$ for $z \notin S$. Given any disk $\overline{D}(z_0, R)$, apply Lemma 4.1 to the entire function Φ with $\theta = N$. If the disk $\overline{D}(z_*, c_N \delta)$ contains a pole z_i of v , then $|\Phi(z_*)| \leq \frac{3}{2} |\Phi(z_i)| \leq \frac{3}{2} M$. If it contains no pole, then $d(z_*) \geq c_N \delta$ and $|\Phi(z_*)| \leq C(1 + (c_N \delta)^{-N})$. In both cases

$$R^N |\Phi(z_0)| \leq \delta^N |\Phi(z_*)| \leq \max(\frac{3}{2} M, C) R^N + C c_N^{-N}.$$

Letting $R \rightarrow \infty$ gives $|\Phi(z_0)| \leq \max(\frac{3}{2} M, C)$ for every z_0 . So Φ is bounded, and constant by Liouville's theorem. $\square$

In the applications Φ takes only finitely many values on S . The hypothesis on the values at the poles cannot be dropped; an example with $p = 2$ is given in Appendix D.

5 From one Laurent series to the global structure

This section does Step 3 of Section 2 in general.

5.1 Equal series give twisted periods

From here on we return to waves. Recall that Γ is the group of twisted periods of a wave, $\chi: \Gamma \rightarrow \mu_m$ its character (Definition 1.1), and S its set of poles. A discrete subgroup of $\mathbb{C}$ is $\{0\}$, a group $\tau\mathbb{Z}$ with $\tau \neq 0$, or a lattice; we say that it has *rank* zero, one or two.

Lemma 5.1 (Equal series and twisted periods). *Let v be a wave.*

- (a) *Two poles z_1, z_2 of v satisfy $z_2 - z_1 \in \Gamma$ if and only if $v(z_2 + z) = \omega v(z_1 + z)$ near $z = 0$ for some $\omega \in \mu_m$.*
- (b) *Γ is a discrete subgroup of $\mathbb{C}$, and the group $\ker \chi$ of ordinary periods has finite index in it.*
- (c) *Suppose the poles of v lie in finitely many cosets of Γ . If $\Gamma = \{0\}$, then v is rational and bounded at infinity. If Γ has rank one and τ generates $\ker \chi$, then v is a rational function of $e^{2\pi iz/\tau}$ that is finite at 0 and at ∞ . If Γ is a lattice, then v is elliptic.*

Proof. (a) If $\gamma = z_2 - z_1 \in \Gamma$, then $v(z_2 + z) = v(z_1 + z + \gamma) = \chi(\gamma)v(z_1 + z)$. Conversely, if the two Laurent series agree up to the factor ω , then $v(z + z_2 - z_1) = \omega v(z)$ near z_1 , hence everywhere by the identity theorem.

(b) Suppose distinct elements $\gamma_i \in \Gamma$ tend to 0. Passing to a subsequence, they have a common multiplier ω . Letting $i \rightarrow \infty$ in $v(z + \gamma_i) = \omega v(z)$ at a point where v is holomorphic and nonzero gives $\omega = 1$. Then $(v(z + \gamma_i) - v(z))/\gamma_i = 0$ gives $v' \equiv 0$, which is impossible. So Γ is discrete. The kernel of χ has finite index because μ_m is finite.

(c) If $\Gamma = \{0\}$, then v has finitely many poles, and Lemma 4.2 applies. If Γ is a lattice, so is $\ker \chi$, and v is elliptic. Suppose Γ has rank one. Each coset of Γ is a finite union of cosets of $\ker \chi = \tau\mathbb{Z}$, so $S = \bigcup_{\nu=1}^N (a_\nu + \tau\mathbb{Z})$. With $y = \text{Im}(z/\tau)$ we have

$$d(z) \geq |\tau| \min_{\nu} |y - \text{Im}(a_\nu/\tau)| \longrightarrow \infty \quad (y \rightarrow \pm\infty),$$

uniformly in $\text{Re}(z/\tau)$. By Lemma 4.2, v is bounded for large $|y|$. Since τ is a period, v is a function of $t = e^{2\pi iz/\tau}$, meromorphic on $\mathbb{C}^*$ with finitely many poles. As $|t| = e^{-2\pi y}$, it is bounded near $t = 0$ and $t = \infty$. These singularities are therefore removable, and v is rational in t . $\square$

All leading coefficients of a wave differ by factors in μ_m . So part (a) says: two poles lie in the same coset of Γ exactly when their normalized series coincide.

Corollary 5.2 (One series, one orbit). (a) *If a wave has the same normalized series at all its poles (for even p , the same free coefficient at all its poles), then it has exactly one pole modulo Γ and belongs to $\mathcal{W}$.*

(b) *If a pole of one wave and a pole of another wave carry the same normalized series, the two waves coincide after a translation and a phase change.*

Proof. (a) By Lemma 5.1(a) all poles lie in one coset of Γ , and part (c) applies. For even p the free coefficient determines the normalized series. (b) If $v_1(z_1 + z) = c\omega_1 f(z)$ and $v_2(z_2 + z) = c\omega_2 f(z)$ near $z = 0$ with the same series f , then $v_2(z) = (\omega_2/\omega_1) v_1(z + z_1 - z_2)$ near z_2 , hence everywhere by the identity theorem. $\square$

5.2 The one-pole spaces

We now describe the functions with one pole modulo Γ , according to the rank of Γ . For $\Gamma = (2\pi i/k)\mathbb{Z}$ we use the functions g_r of Theorem A,

$$g_r(z) = \frac{k e^{rkz/m}}{e^{kz} - 1}, \quad 1 \leq r \leq m, \quad k \neq 0. \quad (5.1)$$

Recall that g_r has simple poles exactly at the points of Γ , with residue 1 at 0, and the multiplier $e^{2\pi ir/m}$ under $z \mapsto z + 2\pi i/k$. When $r < m$, g_r tends to 0 at both ends, while $g_m = G$ tends to k at the end $\operatorname{Re}(kz) \rightarrow +\infty$ and to 0 at the other end. Replacing k by $-k$, which gives the same Γ , turns g_r into g_{m-r} for $r < m$, and g_m into $g_m - k$; so the form (1.4) does not depend on the choice between k and $-k$.

For a lattice Γ , a character $\chi: \Gamma \rightarrow \mathbb{C}^*$ of finite order and $j \geq 0$, let V_j^χ be the space of meromorphic functions w with $w(z + \gamma) = \chi(\gamma)w(z)$ for $\gamma \in \Gamma$, whose poles lie in Γ and have order at most j . These are elliptic functions of the second kind with multipliers of finite order. The following description of them is classical [2, 17]. It makes Theorem A(iii) explicit and is not used in the proofs of the three theorems; its proof, a construction with the Weierstrass sigma function, is in Appendix B.

Lemma 5.3 (Functions of the second kind). *Let χ be a character of finite order of the lattice Γ . If χ is not trivial, then $V_0^\chi = 0$ and $\dim V_j^\chi = j$ for $j \geq 1$; there is a unique $g_\chi \in V_1^\chi$ with residue 1 at 0, and $g_\chi, Dg_\chi, \dots, D^{j-1}g_\chi$ is a basis of V_j^χ . If χ is trivial, then $V_0^\chi = V_1^\chi = \mathbb{C}$, and for $j \geq 2$ a basis of V_j^χ is $1, \wp, D\wp, \dots, D^{j-2}\wp$.*

For $n = 3$ the functions g_χ are, up to scale, Jacobi elliptic functions (Appendix B); the function y of Section 2.6 is one of them.

Proposition 5.4 (The one-pole spaces). *Let v be a wave with exactly one pole modulo Γ , placed at 0. Then v has the form (i), (ii) or (iii) of Theorem A, according to the rank of Γ . In case (iii), $v \in V_q^\chi$.*

Proof. By Lemma 5.1(c), applied with a single coset, v is rational and bounded at infinity, or rational in an exponential and bounded at both ends, or elliptic.

Rank zero. Then 0 is the only pole, so $v = \beta + \sum_{j=1}^q d_j z^{-j}$ with $d_q \neq 0$. Let j_0 be the smallest index with $d_{j_0} \neq 0$. At infinity, the coefficients of z^0 and of z^{-j_0} in $P(D)v + v^n/n$ are $P(0)\beta + \beta^n/n$ and $(P(0) + \beta^m)d_{j_0}$. Both vanish, so $P(0) = -\beta^m$ and then $(\frac{1}{n} - 1)\beta^n = 0$. Hence $\beta = 0$.

Rank one. Let $\Gamma = (2\pi i/k)\mathbb{Z}$ and let $e^{2\pi ir/m}$ be the multiplier of v under $z \mapsto z + 2\pi i/k$. The functions $D^j g_r$, $0 \leq j < q$, have this multiplier, their principal parts at 0 are $(-1)^j j! z^{-j-1}$, and they are bounded at both ends. Choose $d_0, \dots, d_{q-1}$ so that the difference $v - \sum_j d_j D^j g_r$ is holomorphic at 0; then $d_{q-1} \neq 0$, because the pole of v has order q . The difference has the same multiplier as v , so it is holomorphic at every point of Γ , hence entire. It is also bounded at both ends. As a function of $e^{2\pi iz/\tau}$, where τ is an ordinary period, it is therefore rational, without poles on $\mathbb{C}^*$ and bounded at 0 and ∞ , that is, a constant. A nonzero constant has trivial multiplier, so the difference is zero if $r < m$, and it is a constant β if $r = m$.

Rank two. This is the definition of V_q^χ , with the character χ of v . $\square$

By Lemma 5.3, a twisted elliptic wave is thus a combination of $g_\chi, Dg_\chi, \dots, D^{q-1}g_\chi$, in analogy with (1.4), and an untwisted one, which requires $q \geq 2$, is a combination of $1, \wp, D\wp, \dots, D^{q-2}\wp$.

5.3 Odd degree

Let p be odd. By Lemma 3.1 all poles of all waves carry the same normalized series. By Corollary 5.2(a) every wave has exactly one pole modulo Γ , and Proposition 5.4 gives its form; by Corollary 5.2(b) any two waves coincide after a translation and a phase change. This proves Theorems A and B for odd p .

For even p it remains to show that the free coefficient takes the same value at all poles of a wave, and at most two values on all waves of one equation, only one when $n \geq 4$ or P is not even.

6 Equations for the free coefficient

This section does Step 4 of Section 2 for every operator of even degree: it constructs expressions like the Ψ of Section 2.4, which have no poles and whose Taylor coefficients at a pole give equations for h . Throughout, p is even, so $K = (q)_p > 0$ and $nK = (q+1)_p$. Recall that the series $f(z; h)$ of (3.3) is defined for every operator P of even degree, with coefficients that are polynomials in h and in the coefficients of P . We write $[h^k]$ and $[z^j]$ for the coefficient of h^k and of z^j in such a series. Table 3 in Appendix A lists the notation.

6.1 Functions without poles

In Section 2.4 the principal part of $u^2 u'' - \frac{3}{2} u(u')^2$ at a pole was cancelled by a multiple of u'' , that is, by a differential operator applied to u . For a general operator the principal part of u at a pole has q terms, which depend on P , and one cannot expect the pole of a second expression to be cancelled by derivatives of u alone. Instead we apply a differential operator to each of two expressions, so that their principal parts cancel against each other.

For a Laurent series α at 0 with principal part $\sum_{r=1}^R \alpha_{-r} z^{-r}$, define the *principal-part polynomial*

$$\hat{\alpha}(\rho) = \sum_{r=1}^R \alpha_{-r} \frac{(-\rho)^{r-1}}{(r-1)!}.$$

Since $D^{r-1}(1/z) = (-1)^{r-1}(r-1)! z^{-r}$, the principal part of α is $\hat{\alpha}(D)(1/z)$. Differentiation maps principal parts to principal parts and holomorphic functions to holomorphic functions, so for every polynomial L the principal part of $L(D)\alpha$ is $L(D)\hat{\alpha}(D)(1/z)$; in particular $\widehat{D\alpha} = \rho \hat{\alpha}$.

Lemma 6.1 (Cancelling principal parts). *For Laurent series α and β at 0 with finite principal parts, the series $\hat{\alpha}(D)\beta - \hat{\beta}(D)\alpha$ is holomorphic at 0.*

Proof. Its principal part is $(\hat{\alpha}\hat{\beta} - \hat{\beta}\hat{\alpha})(D)(1/z) = 0$. □

For convergent series the lemma has a transparent reason. Let α and β be holomorphic in $0 < |\zeta| < 2\delta$. For $0 < |z| < \delta$ the residues at $\zeta = 0$ and $\zeta = z$ give

$$\hat{\alpha}(D)\beta(z) - \hat{\beta}(D)\alpha(z) = \frac{1}{2\pi i} \oint_{|\zeta|=\delta} \alpha(\zeta) \beta(z - \zeta) d\zeta,$$

and the integral is holomorphic in $|z| < \delta$.

The function Ψ of Section 2.4 arises from the lemma. Take $E = u$ and $F = u^2 u'' - \frac{3}{2} u(u')^2$. On the series $f(z; h)$ of that example, the principal-part polynomials of E and F are $\hat{E}(\rho) = -\rho$ and $\hat{F}(\rho) = \frac{s}{108} \rho^3$, and

$$\hat{E}(D)F - \hat{F}(D)E = -D\left(F + \frac{s}{108} u''\right) = -\Psi'.$$

In general we need two expressions in u with two properties. Their principal parts on the series $f(z; h)$ must not depend on h ; then $\hat{E}$ and $\hat{F}$ are computed from P . And the two expressions must transform in the same way under phase changes; then at a pole with phase ω_i the principal parts of $E(u)$ and $F(u)$ are those of $E(f)$ and $F(f)$ times the same power of ω_i . So one differential polynomial in u is holomorphic at all poles of the wave, and its expansions at the poles differ only by roots of unity and by the value of h . Thus let E and F be polynomials in $u, u', u'', \dots$ with constant coefficients such that:

(H1) for some integer e , $E(\omega u) = \omega^e E(u)$ and $F(\omega u) = \omega^e F(u)$ for all $\omega \in \mu_m$;

(H2) the principal parts at 0 of $E(f(\cdot; h))$ and $F(f(\cdot; h))$ do not depend on h .

Let $\hat{E}$ and $\hat{F}$ be the principal-part polynomials of $E(f)$ and $F(f)$. By (H2) they depend only on P . Put

$$\varphi(z; h) = \hat{E}(D) F(f) - \hat{F}(D) E(f). \quad (6.1)$$

By Lemma 6.1 this is a power series in z , and its coefficients are polynomials in h and in the coefficients of P .

Lemma 6.2. *Assume (H1) and (H2), and let v be a wave. Then the function*

$$\Phi = \hat{E}(D) F(u) - \hat{F}(D) E(u), \quad u = v/c,$$

is entire, and at each pole z_i of v , $\Phi(z_i + z) = \omega_i^e \varphi(z; h_i)$, with ω_i, h_i as in (3.4).

Proof. Near z_i we have $u(z_i + z) = \omega_i f(z; h_i)$. By (H1), $E(u)(z_i + z) = \omega_i^e E(f(\cdot; h_i))$ and similarly for F . Hence $\Phi(z_i + z) = \omega_i^e \varphi(z; h_i)$, which is holomorphic. Away from the poles, Φ is holomorphic because u is. $\square$

By Lemma 6.2 the values of Φ at the poles are $\omega_i^e \varphi(0; h_i)$. If the constant term $\varphi(0; h)$ does not depend on h , these values lie in a finite set, and Lemma 4.4 makes Φ constant. Then the coefficient of every positive power of z in $\varphi(z; h_i)$ vanishes. These are the equations for h_i . In the example, where $\Phi = -\Psi'$, the constant term $\varphi(0; h)$ is zero.

6.2 The choice of the two expressions

It remains to choose E and F . What we need from a pair is a positive power of z whose coefficient in $\varphi(z; h)$ contains h with a nonzero numerical factor: on a wave this coefficient vanishes, and this is an equation for h . Both pairs used below are built from the quadratic expression

$$H(u) = uu'' - \frac{q+1}{q}(u')^2, \quad (6.2)$$

which vanishes for $u = z^{-q}$; its partner is a power of u with the same behaviour under phase changes. For $n \geq 4$ the pair (u^2, H) gives an equation of degree one (Section 6.3). For $n \leq 3$ that computation no longer applies, and we use the pair (u, H_n) with $H_n = u^{n-2}H$. For this pair the coefficients of φ below z^{J-1} have degree at most one in h , and their terms in h can all vanish, as they do for $D^4 - s$; but the coefficient of z^{J-1} always contains h^2 , and this leads to an equation of degree two (Section 6.4). The terms linear in h return in Section 7.3, where they detect the odd powers in P .

Lemma 6.3 (The expression H). For $u = z^{-q}(1 + \varepsilon z^j)$,

$$H(u) = \varepsilon j(j+1) z^{j-2q-2} + O(\varepsilon^2).$$

In particular $H(z^{-q}) = 0$, so $H(f)$ has a pole of order at most $2q+1$. Moreover

$$[h] H(f) = J(J+1) z^{p-2}(1 + O(z)), \quad [h^k] H(f) = O(z^{kJ-2q-2}) \quad (k \geq 2).$$

Proof. Put $y = u^{-1/q}$, so that $u = y^{-q}$. Then $u' = -q y^{-q-1} y'$ and $u'' = q(q+1) y^{-q-2} (y')^2 - q y^{-q-1} y''$, and therefore

$$H(u) = -q y^{-2q-1} y''. \quad (6.3)$$

For $u = z^{-q}(1 + \varepsilon z^j)$ we have $y = z^{-\frac{\varepsilon}{q} z^{j+1} + O(\varepsilon^2)}$ and $y'' = -\frac{\varepsilon}{q} j(j+1) z^{j-1} + O(\varepsilon^2)$, which gives the first formula. The formulas for $H(f)$ follow from it, with $j = J$, and from Lemma 3.2, because H is quadratic with two derivatives in total and $J - 2q - 2 = p - 2$. $\square$

6.3 The case $n \geq 4$: one value

The pair (u^2, H) requires $p > 2q+1$, that is, $(n-3)q > 1$. For even p this is the case exactly when $n \geq 4$: the inequality holds for $n \geq 5$ and for $n = 4, q \geq 2$; it does not hold for $n \leq 3$; and for $n = 4, q = 1$ the degree $p = 3$ is odd.

Proposition 6.4 (One value). Let p be even and $n \geq 4$, so that $p > 2q+1$. Take $E = u^2$ and $F = H$. Then

$$[z^{p-2q-1}] \varphi(z; h) = \Delta_1 (h - h_*), \quad \Delta_1 = -\frac{J(J+1)(p-2)!}{(2q-1)!(p-2q-1)!} \neq 0,$$

for a number h_* depending only on P , and h_* is the free coefficient at every pole of every wave. In particular, every wave has one pole modulo Γ , and any two waves coincide after translation and phase change.

Proof. Hypotheses. Both u^2 and H are multiplied by ω^2 under $u \mapsto \omega u$. By Lemma 3.2 the part of f^2 depending on h is $O(z^{J-2q}) = O(z^p)$, and by Lemma 6.3 that of $H(f)$ is $O(z^{p-2})$. Both are holomorphic, so (H2) holds. The polynomial $\hat{E}$ has degree $2q-1$ and leading coefficient $-1/(2q-1)!$, and $\hat{F}$ has degree at most $2q$.

The dependence on h . In $[h]\varphi = \hat{E}(D)[h]H(f) - \hat{F}(D)[h]f^2$, the term of highest order in $\hat{E}(D)$, namely $-D^{2q-1}/(2q-1)!$, applied to $J(J+1)z^{p-2}$, gives $\Delta_1 z^{p-2q-1}$. All other terms of $\hat{E}(D)[h]H(f)$ contain higher powers of z . Since $[h]f^2 = 2z^p(1 + O(z))$ and $\hat{F}$ has degree at most $2q$, the second term is $O(z^{p-2q})$. Hence

$$[h]\varphi = \Delta_1 z^{p-2q-1} + O(z^{p-2q}).$$

For $k \geq 2$, Lemmas 3.2 and 6.3 give $[h^k]\varphi = O(z^{kJ-4q-1})$, and $kJ - 4q - 1 \geq 2p - 1 > p - 2q - 1$. So the coefficient of z^{p-2q-1} in φ is affine in h with slope $\Delta_1 \neq 0$. Because $p - 2q - 1 > 0$, the constant term $\varphi(0)$ of φ does not depend on h .

Conclusion. By Lemma 6.2, Φ is entire and its values at the poles lie in the finite set $\{\omega^2 \varphi(0) : \omega \in \mu_m\}$. By Lemma 4.4, Φ is constant. The coefficient of z^{p-2q-1} in $\Phi(z_i + z)$, which is $\omega_i^2 \Delta_1 (h_i - h_*)$, therefore vanishes, and $h_i = h_*$. The last assertion follows from Corollary 5.2. $\square$

6.4 The cases $n = 2$ and $n = 3$: at most two values

For $n \leq 3$ we have $p \leq 2q$; then D^{2q-1} annihilates z^{p-2} , and the term Δ_1 disappears. We take instead $E = u$ and $F = H_n$, where

$$H_n(u) = u^{n-2}H(u) = u^{n-1}u'' - \frac{q+1}{q}u^{n-2}(u')^2,$$

and obtain an equation of degree two (Proposition 6.6). The proof uses an elementary inequality for $n = 2$, which we state first.

Lemma 6.5. *For every integer $q \geq 4$, the number $X = (4q+1)_q/(q)_q$ satisfies $X > 6q+3$.*

Proof. Each of the q factors $(4q+1+i)/(q+i)$, $0 \leq i < q$, of X is at least $5q/(2q-1) > 5/2$, so $X > (5/2)^q$. Moreover $(5/2)^q > 6q+3$ for $q \geq 4$: this holds for $q = 4$, and $(5/2)^{q+1} - (6q+9) > \frac{5}{2}(6q+3) - (6q+9) = 9q - \frac{3}{2} > 0$. $\square$

Proposition 6.6 (A quadratic equation). *Let $p \geq 4$ be even and $q \geq 2$; this covers every even $p \geq 4$ with $n = 2$ or $n = 3$. There is a polynomial Q_P of degree two, depending only on P , such that $Q_P(h) = 0$ for the free coefficient h at every pole of every wave.*

Proof. Both u and H_n are multiplied by ω under $u \mapsto \omega u$, since $\omega^n = \omega$ for $\omega \in \mu_m$; so (H1) holds with $e = 1$. Let $a_* = (-1)^{q-1}/(q-1)!$ be the leading coefficient of $\widehat{E}$. Write $f = f_0 + hf_1 + h^2f_2 + O(h^3)$. By Lemma 3.2,

$$f_0 = z^{-q}(1 + O(z)), \quad f_1 = z^{nq}(1 + O(z)), \quad f_2 = \psi z^{(2n+1)q}(1 + O(z)),$$

where ψ is the coefficient of h^2 in b_{2J} . By the same lemma a monomial of b_{2J} that contains h^2 has weight $2J$, so it is a multiple of h^2 alone. In the recurrence at the index $2J$ such a term arises only from the $\binom{n}{2}$ products $b_J b_J b_0^{n-2}$ in u^n . So

$$\psi := [h^2] b_{2J} = \binom{n}{2} \frac{K}{I(2J)} = \frac{\binom{n}{2}}{X - n}, \quad X = \frac{((n+2)q+1)_p}{(q)_p}; \quad (6.4)$$

here we used that the product $(2J-q)(2J-q-1) \cdots (2J-q-p+1)$ in $I(2J)$ equals $((n+2)q+1)_p$. Since $((n+2)q+1)_p > (q+1)_p = n(q)_p$, we have $X > n$ and $\psi > 0$.

The proof has five parts. The first three show that the terms of φ of degree at least two in h begin with $\Delta_2 h^2 z^{J-1}$, where $\Delta_2 \neq 0$; the last two derive the quadratic equation from this.

Hypothesis (H2). The part of $E(f) = f$ depending on h is $O(z^{nq})$, hence holomorphic. The expression H_n is a product of n factors with two derivatives in total; by Lemma 3.2, its part of degree $k \geq 1$ in h is $O(z^{kJ-nq-2})$, and for $k = 1$ the exponent is $q-2 \geq 0$. So (H2) holds. The polynomial $\widehat{E}$ has degree $q-1$, and $\widehat{F}$ has degree at most nq because $H_n(f)$ has a pole of order at most $(n-2)q+2q+1$.

The terms of $H_n(f)$ in h and h^2 . Let $\mathcal{B}$ be the symmetric bilinear form with $\mathcal{B}(g, g) = H(g)$. Then $[h]H(f) = 2\mathcal{B}(f_0, f_1)$ and $[h^2]H(f) = 2\mathcal{B}(f_0, f_2) + H(f_1)$, and Lemma 6.3 gives

$$2\mathcal{B}(f_0, f_1) = J(J+1)z^{p-2}(1 + O(z)), \quad 2\mathcal{B}(f_0, f_2) = 2J(2J+1)\psi z^{2nq-2}(1 + O(z)),$$

while $H(f_1) = (nq(nq-1) - \frac{q+1}{q}(nq)^2)z^{2nq-2}(1 + O(z)) = -nJ z^{2nq-2}(1 + O(z))$. Recall that $H(f_0) = O(z^{-2q-1})$. For the term linear in h ,

$$[h]H_n(f) = f_0^{n-2} \cdot 2\mathcal{B}(f_0, f_1) + (n-2)f_0^{n-3}f_1 H(f_0) = J(J+1)z^{q-2}(1 + O(z)),$$

because the second term is $O(z^{q-1})$. For the quadratic term,

$$\begin{aligned} [h^2] H_n(f) &= f_0^{n-2} (2\mathcal{B}(f_0, f_2) + H(f_1)) + (n-2)f_0^{n-3} f_1 \cdot 2\mathcal{B}(f_0, f_1) \\ &\quad + \left((n-2)f_0^{n-3} f_2 + \binom{n-2}{2} f_0^{n-4} f_1^2 \right) H(f_0). \end{aligned}$$

The first two terms begin with the power $z^{(n+2)q-2}$, with the coefficients $2J(2J+1)\psi - nJ$ and $(n-2)J(J+1)$. The last term is $O(z^{(n+2)q-1})$, again because the leading pole of $H(f_0)$ cancels. Altogether,

$$[h^2] H_n(f) = \kappa z^{(n+2)q-2} (1 + O(z)), \quad \kappa = J(2(2J+1)\psi + (n-2)J - 2). \quad (6.5)$$

For $n \geq 3$ we have $\kappa > 0$, since $\psi > 0$ and $(n-2)J - 2 > 0$. For $n = 2$ the degree $p = q$ is even and at least 4, $J = 3q$ and $\psi = 1/(X-2)$ with $X = (4q+1)_q/(q)_q$, so that

$$\kappa = 6q \left(\frac{6q+1}{X-2} - 1 \right) < 0$$

by Lemma 6.5. In all cases $\kappa \neq 0$.

The terms of φ of degree at least two in h . Only $a_* D^{q-1} H_n(f)$ contributes to the lowest h^2 -term of φ :

$$[h^2] \varphi = \Delta_2 z^{J-1} + O(z^J), \quad \Delta_2 = a_* \kappa \frac{((n+2)q-2)!}{(J-1)!} \neq 0. \quad (6.6)$$

Indeed, $D^{q-1} z^{(n+2)q-2} = \frac{((n+2)q-2)!}{(J-1)!} z^{J-1}$, lower derivatives of $H_n(f)$ give higher powers of z , and since $[h^2]f = O(z^{2J-q})$ and $\hat{F}$ has degree at most nq , the term $\hat{F}(D)[h^2]f$ is $O(z^{2J-q-nq}) = O(z^J)$. In the same way, $[h^k]\varphi = O(z^{(k-1)J-1})$ for $k \geq 3$. So the terms of φ of degree at least two in h start at z^{J-1} , and

$$[z^i] \varphi = A_i h + B_i \quad (0 \leq i \leq J-2), \quad (6.7)$$

where A_i and B_i are polynomials in the coefficients $a_0, \dots, a_{p-1}$ of P . In summary, the coefficients of h^k in f , $H_n(f)$ and φ begin as follows; an entry without O is a leading term, exact up to a factor $1 + O(z)$, and φ is a power series by Lemma 6.1:

k	f	$H_n(f)$	φ
0	z^{-q}	$O(z^{-nq-1})$	$O(1)$
1	z^{nq}	$J(J+1)z^{q-2}$	$O(1)$
2	$\psi z^{(2n+1)q}$	$\kappa z^{(n+2)q-2}$	$\Delta_2 z^{J-1}$
$k \geq 3$	$O(z^{kJ-q})$	$O(z^{kJ-nq-2})$	$O(z^{(k-1)J-1})$

(6.8)

A function whose values at the poles do not depend on h . Let Φ be the entire function of Lemma 6.2. At a pole z_i we have $\Phi(z_i) = \omega_i(A_0 h_i + B_0)$ and, since $J-2 \geq 1$, $\Phi'(z_i) = \omega_i(A_1 h_i + B_1)$. Put

$$T = \Phi \text{ if } A_0 = 0, \quad T = A_0 \Phi' - A_1 \Phi \text{ if } A_0 \neq 0.$$

Then T is entire, and its value at a pole z_i is $\omega_i B_0$ in the first case and $\omega_i(A_0 B_1 - A_1 B_0)$ in the second. These values lie in a finite set, so T is constant by Lemma 4.4. (By Lemma 7.2(b) below, the first case is $a_{p-1} = 0$; this is not used in the proof.)

The quadratic equation. If $A_0 = 0$, the coefficient of z^{J-1} in $T(z_i + z)$ is ω_i times a polynomial in h_i of degree two with leading coefficient Δ_2 . If $A_0 \neq 0$, the coefficient of z^{J-2} in $T(z_i + z)$ is ω_i times a polynomial in h_i of degree two with leading coefficient $A_0(J-1)\Delta_2$: the derivative Φ' contributes $(J-1)\Delta_2 h_i^2$ at this power, while Φ has no h^2 -term below z^{J-1} . In both cases the power of z is positive, so the coefficient vanishes because T is constant. Dividing by ω_i gives the polynomial Q_P . $\square$

In Section 2.4, $A_0 = 0$, $T = \Phi = -\Psi'$, and Q_P is the coefficient of z^7 in $-\Psi'$. Proposition 6.6 does not extend to $p = 2$, where one equation can have infinitely many waves (Appendix D): at $(n, q) = (2, 2)$ one finds $X = 15$ and $\kappa = 0$.

7 Proof of Theorem A and of the bounds in Theorem B

This section does Step 5 of Section 2 in general and proves Theorem A and the bounds in Theorem B.

7.1 Two values in one wave

For $n = 2, 3$ a wave might carry the two roots of Q_P at different poles. A count of zeros and poles shows that this does not happen.

Lemma 7.1 (Two values, one orbit). *Let p be even, and let v be a wave whose free coefficient takes at most two values at its poles. Then v has exactly one pole modulo Γ .*

Proof. The free coefficient determines the normalized series, so the series of v fall into at most two classes. By Lemma 5.1(a) its poles lie in at most two cosets of Γ , and by part (c) it lies in $\mathcal{W}$. Suppose there are two, containing poles z_1 and z_2 , and write $v(z_i + z) = c\omega_i f(z; h_i)$ as in (3.4). Put

$$a = z_2 - z_1, \quad \omega = \frac{\omega_2}{\omega_1} \in \mu_m, \quad W(x) = v(z_1 + a + x) - \omega v(z_1 + x).$$

We show that $W \equiv 0$. Then $a \in \Gamma$, which contradicts the choice of z_1 and z_2 . Suppose $W \not\equiv 0$.

Zeros and poles of W . By Corollary 3.3(b), W has a zero of order at least nq at $x = 0$. It has the same character as v , so the same holds at every point of Γ . The poles of v form the set $(z_1 + \Gamma) \cup (z_1 + a + \Gamma)$, so the poles of W lie in $\Gamma \cup (a + \Gamma) \cup (-a + \Gamma)$. On Γ the function W vanishes, so its poles lie in the two cosets $\pm a + \Gamma$, and they have order at most q .

The compact surface. Let $\overline{\mathbb{C}/\Gamma}$ be the Riemann sphere $\mathbb{C} \cup \{\infty\}$ if $\Gamma = \{0\}$; the cylinder $\mathbb{C}/\Gamma$ with its two ends added, again a sphere, if Γ has rank one; and the torus $\mathbb{C}/\Gamma$ if Γ is a lattice. The function W^m is invariant under Γ , so it is a meromorphic function on $\mathbb{C}/\Gamma$. By Lemma 5.1(c), v and hence W^m are bounded near the added points, so W^m extends to a meromorphic function on $\overline{\mathbb{C}/\Gamma}$. The quotient map is locally biholomorphic, so it preserves orders of zeros and poles. A nonzero meromorphic function on a compact Riemann surface has as many zeros as poles, counted with multiplicity.

The case $n \geq 3$. On $\overline{\mathbb{C}/\Gamma}$ the function W^m has poles of total order at most $2qm$, and a zero of order at least nqm . This is a contradiction.

The case $n = 2$. Now $m = 1$ and $\omega = 1$, the leading coefficient c is the same at all poles, and $q = p$ is even. If $2a \in \Gamma$, the two cosets $\pm a + \Gamma$ coincide; then W has poles of total order at most q and a zero of order at least $2q$, a contradiction. If $2a \notin \Gamma$, this count gives no contradiction: W

has poles of total order at most $2q$ and a zero of order at least $2q$. One more zero comes from the odd part of W . Consider

$$\Omega(x) = W(x) - W(-x) = v(z_1 + a + x) - v(z_1 + x) - v(z_1 + a - x) + v(z_1 - x).$$

The reflection $x \mapsto -x$ acts on $\overline{\mathbb{C}/\Gamma}$, so Ω is meromorphic there, with poles only at the two points $\pm a$, of order at most q . It is odd and $O(x^{2q})$ at 0, hence $O(x^{2q+1})$. At $x = -a$ the four terms are evaluated at z_1 , $z_1 - a$, $z_1 + 2a$ and $z_1 + a$. The points $z_1 - a$ and $z_1 + 2a$ are not poles of v , since neither a nor $2a$ lies in Γ . So near $x = -a$ only the term $v(z_1 + a + x)$ of $W(x)$ has a pole, with leading coefficient c , and only the term $-v(z_1 - x)$ of $W(-x)$ has a pole, with leading coefficient $-c$ because q is even. Thus $\Omega(x) = 2c(x + a)^{-q} + \dots$ is not identically zero, and on $\overline{\mathbb{C}/\Gamma}$ it has at least $2q + 1$ zeros and at most $2q$ poles. This is impossible. So $W \equiv 0$ in all cases. $\square$

7.2 Proof of Theorem A, and the bound of two waves

Let v be a wave. By Proposition 4.3 it has a pole, so $q = p/m$ is a positive integer. For odd p the theorem was proved in Section 5.3. Let p be even. If $n \geq 4$, Proposition 6.4 shows that all poles of all waves carry the same free coefficient, and Corollary 5.2(a) applies. If $n \in \{2, 3\}$, then $p \geq 4$, because $p \geq 3$ is even, and $q \geq 2$; by Proposition 6.6 the free coefficients at the poles of all waves are roots of Q_P , and Lemma 7.1 applies. In each case v has exactly one pole modulo Γ , and Proposition 5.4 gives its form. This proves Theorem A.

By Corollary 5.2(b), two waves that share a value of the free coefficient coincide after translation and phase change. So an equation has at most as many waves as there are values of the free coefficient: at most two, and only one if $n \geq 4$. This proves the first assertion of Theorem B.

7.3 Operators that are not even

For $n = 2, 3$ the quadratic equation leaves two possible values of the free coefficient. The terms of φ that are linear in h decide between them whenever P is not even. We keep the pair $E = u$, $F = H_n$ and the assumptions $p \geq 4$ even and $q \geq 2$, and recall from (6.7) that $[z^i]\varphi = A_i h + B_i$ for $0 \leq i \leq J - 2$, and that a_{p-k} has weight k .

Lemma 7.2 (The terms linear in h). *Let $p \geq 4$ be even and $q \geq 2$.*

(a) A_i is a weighted-homogeneous polynomial of weight $i + 1$.

(b) $A_0 = c_0 a_{p-1}$ and $A_1 = c_1 a_{p-1}^2$, where

$$c_0 = (-1)^q \frac{2(n+1)(nq+1)(J+1)}{(n-1)(J-1)}, \quad c_1 = -\frac{nq+2}{(n-1)(J-1)} c_0.$$

(c) Let $n \in \{2, 3\}$, let j be odd with $3 \leq j \leq p - 1$, and suppose that $a_{p-i} = 0$ for all odd $i < j$. Then $A_{j-1} = c_{j-1} a_{p-j}$ with a number $c_{j-1} \neq 0$ depending only on n , q and j .

Proof. (a) By the proof of Lemma 3.2, $\lambda^q f(\lambda z; h)$ is the series (3.3) of the operator P_λ with free coefficient $\lambda^J h$. The expressions $E = u$ and $F = H_n$ are homogeneous: H_n has n factors and two derivatives, so $H_n(\lambda^q f(\lambda \cdot))(z) = \lambda^{nq+2} H_n(f)(\lambda z)$. If a Laurent series α has the principal-part polynomial $\hat{\alpha}(\rho)$, then $\lambda^N \alpha(\lambda z)$ has $\lambda^{N-1} \hat{\alpha}(\rho/\lambda)$, for every integer N . Inserting this into (6.1) shows that the series φ of P_λ , taken at the free coefficient $\lambda^J h$, equals $\lambda^{J+1} \varphi(\lambda z; h)$. Comparing the coefficients of $h z^i$ gives $A_i(P_\lambda) = \lambda^{i+1} A_i(P)$, and the coefficients of P_λ are $\lambda^k a_{p-k}$.

(b), (c) By (a), A_0 is a multiple of a_{p-1} , and A_1 is a combination of a_{p-1}^2 and a_{p-2} . Under the hypothesis of (c), A_{j-1} has odd weight j , so each of its monomials contains a coefficient a_{p-i} with i odd; those with $i < j$ vanish, and the only monomial of weight j containing a_{p-i} with $i \geq j$ is a_{p-j} itself. It remains to find the numerical coefficients. They depend only on n, q and j . Since $[z^i]\varphi = A_i h + B_i$, the number A_i is the coefficient of z^i in $\partial\varphi/\partial h$ at every value of h . So the coefficients can be computed from any convenient operator and any value of h . This is done in Appendix C by three explicit computations with Laurent series. First, $A_1 = 0$ for every even operator, so A_1 contains no term a_{p-2} . Second, c_0 and c_1 are read off from an operator that has an explicit wave; at the free coefficient of this wave the derivative of $f(z; h)$ with respect to h is also explicit. Third, c_{j-1} is found from the operator $D^p + a D^{p-j}$ to first order in a , and an inequality between binomial coefficients shows that it is not zero. $\square$

Proposition 7.3 (One wave unless P is even). *Let $p \geq 4$ be even and $n \in \{2, 3\}$.*

- (a) *If $A_i \neq 0$ for some i with $1 \leq i \leq J-2$, then the free coefficient equals $-B_i/A_i$ at all poles of all waves, and (1.1) has at most one wave up to translation and phase change.*
- (b) *If P is not even, then $A_i \neq 0$ for some i with $1 \leq i \leq p-2$.*

Proof. (a) Let v be a wave, and let Φ be the entire function of Lemma 6.2 for $E = u, F = H_n$. By Proposition 6.6 the free coefficients h_i at the poles of v are roots of Q_P . So the values $\omega_i(A_0 h_i + B_0)$ of Φ at the poles lie in a finite set, and Φ is constant by Lemma 4.4. Hence the coefficients of all positive powers of z in $\varphi(z; h)$ vanish for the free coefficient h at any pole of any wave; by (6.7),

$$A_i h + B_i = 0 \quad (1 \leq i \leq J-2).$$

If $A_i \neq 0$ for one of these i , then $h = -B_i/A_i$. So all poles of all waves carry the same normalized series, and Corollary 5.2 applies.

(b) Since P is not even and p is even, there is a least odd j with $a_{p-j} \neq 0$, and $j \leq p-1$. If $j = 1$, then $A_1 = c_1 a_{p-1}^2 \neq 0$. If $j \geq 3$, then $A_{j-1} = c_{j-1} a_{p-j} \neq 0$, and $2 \leq j-1 \leq p-2$. $\square$

So an equation has at most one wave if $n \in \{2, 3\}$ and P is not even. Together with Section 5.3 and Proposition 6.4 this proves the second assertion of Theorem B; Table 1 collects the cases. The last assertion, that $D^4 - s$ with $s \neq 0$ has exactly two waves, is proved in Section 2.6 for $n = 3$ and in Section 8.3 for $n = 2$.

Proposition 7.3(a) can also apply to an even operator when $p \geq 6$. For $n = 3$ and $P(D) = D^6 + a_0$ with $a_0 \neq 0$ one finds $A_5 = -\frac{13}{888} a_0$, so the equation $v^{(6)} + a_0 v + v^3/3 = 0$ has only one wave, $v = -\frac{c}{2} \varphi'(z; 0, -a_0/4320)$.

case	values of h on all waves of one equation	proved in
p odd	no free coefficient	Lemma 3.1
p even, $n \geq 4$	at most one	Proposition 6.4
p even, $n = 2, 3$, P not even	at most one	Propositions 6.6 and 7.3
p even, $n = 2, 3$, P even	at most two	Proposition 6.6

Table 1: The free coefficient h in the proof of Theorems A and B. In each case a wave has one pole modulo Γ (Corollary 5.2(a) in the first two rows, Lemma 7.1 in the last two) and therefore carries one value of h (Lemma 5.1(a)); each value belongs to at most one wave (Corollary 5.2(b)).

8 The power of a wave

This section does Step 7 of Section 2 in general and proves Theorem C.

8.1 The formula

Let $p \geq 3$ and let $u = z^{-q} \sum_j b_j z^j$ be a normalized wave. The coefficients $b_1, \dots, b_p$ precede the only possible resonance $J = p + 2q$, so they are determined by P , and the recurrence defines them for every operator, whether or not a wave exists. Define $\ell_0 = 1, \ell_1, \dots, \ell_p$ by

$$\left(\sum_{i=0}^p b_i(P) z^i \right)^m = \sum_{j=0}^p \ell_j(P) z^j + O(z^{p+1}). \quad (8.1)$$

Then every normalized wave satisfies

$$u^m = \sum_{j=0}^p \ell_j(P) z^{j-p} + O(z),$$

so both the principal part and the constant term of u^m are fixed by P .

The power u^m has trivial character, because χ takes values in μ_m ; so u^m is invariant under Γ , with one pole of order p modulo Γ . We shall write it as a combination of standard functions Λ_r . For each group Γ , and for r in the range given below, there is a unique function Λ_r with the following properties. It is invariant under Γ , has poles only on Γ , and satisfies $\Lambda_r = z^{-r} + O(z)$ at 0. It is bounded at infinity if $\Gamma = \{0\}$, and at the two ends if Γ has rank one. Explicitly:

- if $\Gamma = \{0\}$: $\Lambda_r = z^{-r}$, $1 \leq r \leq p$;
- if $\Gamma = (2\pi i/k)\mathbb{Z}$: with $G = k e^{kz} / (e^{kz} - 1)$,

$$\Lambda_r = \frac{(-1)^{r-1}}{(r-1)!} \left(D^{r-1} G - [z^0] D^{r-1} G \right), \quad 1 \leq r \leq p;$$

- if Γ is a lattice: with $\wp = \wp(z; \Gamma)$,

$$\Lambda_r = \frac{(-1)^r}{(r-1)!} \left(D^{r-2} \wp - [z^0] D^{r-2} \wp \right), \quad 2 \leq r \leq p.$$

Since $D^j G = (-1)^j j! z^{-j-1} + O(1)$ and $D^j \wp = (-1)^j (j+1)! z^{-j-2} + O(1)$, these functions indeed satisfy $\Lambda_r = z^{-r} + O(z)$. Each Λ_r is unique, because the difference of two candidates is a bounded function without poles on $\mathbb{C}/\Gamma$ (or on the sphere), hence a constant, and its constant term is zero. No elliptic function has a single simple pole, so there is no Λ_1 in the lattice case. Writing a function of $\mathcal{W}$ through the derivatives of one function, here G or $\wp$, is Hermite's decomposition, a standard tool for constructing meromorphic travelling waves [10–12, 31].

Theorem 8.1 (The formula for u^m). *Let $p \geq 3$ and let u be a normalized wave with group of twisted periods Γ . Then*

$$u^m = \ell_p + \sum_{r=1}^p \ell_{p-r} \Lambda_r, \quad (8.2)$$

with the functions Λ_r of the group Γ , and:

- (i) if $\Gamma = \{0\}$, then $\ell_p = 0$;
- (ii) if $\Gamma = (2\pi i/k)\mathbb{Z}$, then the right-hand side is a monic polynomial of degree p in G ; the residue ℓ_{p-1} may be nonzero (Corollary 8.2);
- (iii) if Γ is a lattice, then $\ell_{p-1} = 0$, the term $r = 1$ is absent, and the right-hand side is an elliptic function for Γ .

In each case the right-hand side is the only function that is invariant under Γ , has poles only on Γ , is bounded at infinity or at the two ends when Γ has rank zero or one, and has the principal part and constant term prescribed by (8.1). In particular, u^m does not depend on the character or on any further coefficient of the wave.

Proof. By Theorem A, u^m is invariant under Γ , has a single pole of order p modulo Γ , and is bounded at infinity or at the two ends when Γ has rank zero or one.

(i) The difference $u^m - \sum_r \ell_{p-r} z^{-r}$ has no poles and is bounded, so it is a constant, equal to its constant term ℓ_p . Both u^m and the sum tend to 0 at infinity by Theorem A(i), so $\ell_p = 0$.

(ii) By (8.1), u^m and the right-hand side of (8.2) have the same principal part and constant term at 0. Their difference is invariant under Γ , has no poles and is bounded at both ends; so it is a constant, which is zero. Since $DG = G(k - G)$, the function $D^{r-1}G$ is a polynomial of degree r in G with leading coefficient $(-1)^{r-1}(r-1)!$; so Λ_p is monic of degree p in G , and $\ell_0 = 1$.

(iii) The residue ℓ_{p-1} of the elliptic function u^m at its only pole in a period cell vanishes. As in (ii), the difference of u^m and the right-hand side is then an elliptic function without poles, hence a constant, which is zero.

Uniqueness, among the functions that are invariant under Γ , have poles only on Γ and are bounded at infinity or at the ends, follows as for Λ_r . $\square$

All coefficients in these formulas are explicit. For instance, $[z^0]D^j G = B_{j+1}(1)k^{j+1}/(j+1)$ with the Bernoulli polynomials B_j , and the constant terms of $D^j \wp$ are polynomials in the invariants of the lattice. The theorem describes u^m once Γ is known; it does not determine the rate k or the lattice.

Corollary 8.2 (Power residue). *Let $p \geq 3$. The number $\mathcal{R}(P) = \ell_{p-1}(P)$ is a polynomial in the coefficients of P , and for a normalized wave it equals $\text{Res}_{z=0} u^m$. Every elliptic wave requires $\mathcal{R}(P) = 0$. For a wave with $\Gamma = (2\pi i/k)\mathbb{Z}$,*

$$u_+^m - u_-^m = k \mathcal{R}(P), \quad (8.3)$$

where $u_{\pm}$ are the limits of u as $\text{Re}(kz) \rightarrow \pm\infty$.

Proof. The elliptic case is part (iii) of Theorem 8.1. If Γ has rank one, then $G \rightarrow k$ as $\text{Re}(kz) \rightarrow +\infty$ and $G \rightarrow 0$ as $\text{Re}(kz) \rightarrow -\infty$. Since $DG = G(k - G)$, every derivative $D^j G$, $j \geq 1$, vanishes at both ends, so Λ_r takes the same value at both ends for $r \geq 2$, while $\Lambda_1 = G - k/2$ increases by k . Formula (8.2) gives (8.3). $\square$

For elliptic solutions, conditions obtained from the vanishing of a sum of residues are a standard tool [4, 18]; the corollary applies it to u^m , which has one pole modulo Γ . Theorem 8.1 and Corollary 8.2 are the precise form of Theorem C, which is thereby proved. Formula (8.2) has further consequences.

Corollary 8.3. *Let $p \geq 3$.*

- (a) *A rational wave has $\ell_{p-r} = 0$ for $1 \leq r < m$, besides $\ell_p = 0$.*
- (b) *A twisted wave with Γ of rank one has $\mathcal{R}(P) = 0$.*
- (c) *If $n \geq 3$ and $\mathcal{R}(P) \neq 0$, every wave is an untwisted front: Γ has rank one, the character is trivial, and the limits at the two ends are different.*
- (d) *Two waves of one equation that are not related by translation and phase change never have the same twisted periods.*

Proof. (a) By Theorem 8.1(i), $u^m = \sum_r \ell_{p-r} z^{-r}$, and $u^m = O(z^{-m})$ at infinity by Theorem A(i).

(b) Such a wave tends to 0 at both ends (Theorem A(ii)), so $\mathcal{R}(P) = 0$ by (8.3).

(c) Now $m \geq 2$ and $\ell_{p-1} = \mathcal{R}(P) \neq 0$. By (a) with $r = 1$ no wave is rational, by Corollary 8.2 none is elliptic, and by (b) none is twisted. So Γ has rank one and the character is trivial, and $u_+^m \neq u_-^m$ by (8.3).

(d) If two normalized waves of the same equation have the same group Γ , then (8.2) gives them the same power u^m ; their quotient is then a constant in μ_m , which is 1 because both begin with z^{-q} . $\square$

By part (d), a wave is determined, up to a phase change, by P , the position of one pole and the group Γ . If Γ has rank one and $\mathcal{R}(P) \neq 0$, the limits satisfy $P(0)u_{\pm} = Ku_{\pm}^n$ by (3.1), so $u_{\pm}^m \in \{0, P(0)/K\}$, and (8.3) gives the rate $k = \pm P(0)/(K\mathcal{R}(P))$.

8.2 The residue condition: the Kuramoto–Sivashinsky equation

The Kuramoto–Sivashinsky equation with dispersion, $U_t + UU_x + \mu U_{xx} + \delta U_{xxx} + \nu U_{xxxx} = 0$, $\nu \neq 0$, also describes pressure waves in a liquid with gas bubbles [24]. Its travelling waves $U = w(x - st)$ satisfy

$$\nu w''' + \delta w'' + \mu w' - sw + \frac{w^2}{2} + C = 0$$

for a constant C . Since $n = 2$, the shift $w = v + w_0$ with $w_0^2/2 - sw_0 + C = 0$ removes C and changes only the constant term of the linear part: v satisfies $P(D)v + v^2/2 = 0$ for the operator $P(D) = \nu D^3 + \delta D^2 + \mu D + a_0$ with $a_0 = w_0 - s$, and $q = 3$.

Consider first $\nu = 1$, so that P is monic. Then $c = 120$, and $u = v/120$ satisfies $P(D)u + 60u^2 = 0$. For $u = z^{-3} + b_1 z^{-2} + b_2 z^{-1} + O(1)$, the coefficients of z^{-5} and z^{-4} give

$$96b_1 + 12\delta = 0, \quad 114b_2 + 6\delta b_1 - 3\mu + 60b_1^2 = 0,$$

so that

$$u = z^{-3} - \frac{\delta}{8} z^{-2} + \frac{16\mu - \delta^2}{608} z^{-1} + O(1).$$

Here $m = 1$ and $\mathcal{R}(P) = \text{Res}_{z=0} u = (16\mu - \delta^2)/608$. For general ν , the function v/ν solves the equation of the monic operator P/ν . So $v = 120\nu u$, where u is a normalized wave of P/ν , and $\mathcal{R} = (16\mu\nu - \delta^2)/(608\nu^2)$. By Corollary 8.2, every elliptic travelling wave requires

$$\delta^2 = 16\mu\nu,$$

independently of s and C [14, Eq. (3) and Theorem 1(i)].

The condition holds for $\delta = 4$ and $\mu = \nu = 1$. At the speed $s = 0$, the functions

$$w = -60 \wp' - 60 \wp - 1, \quad \wp = \wp(z; \frac{1}{12}, g_3),$$

solve $w''' + 4w'' + w' + w^2/2 + C = 0$ with $C = 1080g_3 - 13$. For $27g_3^2 \neq 12^{-3}$ they are elliptic. So at a single speed there are infinitely many elliptic travelling waves, one for each integration constant other than -8 and -18 . This is why the bound of Theorem B refers to a fixed integration constant as well as to a fixed speed: the operator P depends on both. For $C = -8$ and $C = -18$, that is, $g_3 = \pm 1/216$, the function $\wp$ degenerates, and the formula gives a wave that is rational in an exponential. By Theorem B it is the only wave, so for these two constants there is no elliptic wave. Thus the condition $\delta^2 = 16\mu\nu$ is necessary for an elliptic wave and not sufficient. The elliptic travelling waves of the Kuramoto–Sivashinsky equation with dispersion were found by Kudryashov [20], and by Eremenko [14] there are no others.

8.3 Two waves for $n = 2$

The bound of two waves is also attained for $n = 2$, and Theorem 8.1 finds the waves. For $P(D) = D^4 - s$ with $s \neq 0$ we have $q = 4$, $m = 1$ and $K = 840$, and the normalized equation is $u'''' = s u + 840 u^2$. The recurrence gives $u = z^{-4} - \frac{s}{1680} + O(z^4)$, so that $\ell_1 = \ell_2 = \ell_3 = 0$ and $\ell_4 = -s/1680$ in (8.1). By Theorem 8.1, every normalized wave is $u = \Lambda_4 - s/1680$. To write Λ_4 in the same way for all three kinds of Γ , we use the Weierstrass function $\wp = \wp(z; g_2, g_3)$ and its degenerations. If Γ is a lattice, $\wp$ is its Weierstrass function. If $\Gamma = (2\pi i/k)\mathbb{Z}$, put $\wp = k^2/12 - DG$ and $(g_2, g_3) = (k^4/12, -k^6/216)$. If $\Gamma = \{0\}$, put $\wp = z^{-2}$ and $(g_2, g_3) = (0, 0)$. In all three cases $\wp'' = 6\wp^2 - g_2/2$ and $\Lambda_4 = \frac{1}{6}(\wp'' - [z^0]\wp'') = \wp^2 - g_2/10$, so that

$$u = \wp^2 - \frac{g_2}{10} - \frac{s}{1680}.$$

Substituting into the equation gives

$$g_3 = 0, \quad g_2^2 = \frac{s^2}{23184}.$$

So $g_3 = 0$ and $g_2 \neq 0$, and $\wp$ is not degenerate. Hence $v'''' - sv + v^2/2 = 0$ has exactly two waves for every $s \neq 0$, both elliptic, one for each root $g_2 = \pm s/(12\sqrt{161})$; they are different waves, because their free coefficients $h = g_2^3/10400$ are opposite and not zero. They are the solutions (26) of Demina and Kudryashov [10], with g_2 a root of the quadratic equation (25) of that paper, at $\beta = 0$ and $C_1 = 0$ in its notation. With Section 2.6, this completes the proof of Theorem B.

The same computation shows that the bound is attained for $p = 6$. For $P(D) = D^6 + a_0$ with $a_0 \neq 0$ it gives $u = \wp^3 - \frac{3g_2}{20}\wp - \frac{3g_3}{28} + \frac{a_0}{665280}$ with $g_2 = 0$ and $g_3^2 = a_0^2/3643833600$. So $v^{(6)} + a_0 v + v^2/2 = 0$ has exactly two waves, both elliptic, one for each sign of g_3 ; they are different waves, because their lattices are different.

9 Further consequences, and an example

9.1 Equations without dissipation

If v solves (1.1), then $v(-z)$, multiplied by a number whose m th power is $(-1)^p$, solves the same equation with $P(\rho)$ replaced by $(-1)^p P(-\rho)$. In particular, for even P the reflection $v(z) \mapsto v(-z)$ is a symmetry of (1.1), and every wave is symmetric:

Proposition 9.1 (Reflection). *Let P be even. Every wave v has one pole modulo Γ , its character has order at most two, and at every pole z_0*

$$v(z_0 + z) = (-1)^q v(z_0 - z). \quad (9.1)$$

Proof. The pole orbit comes from Theorem A if $p \geq 3$ and from Proposition D.1 if $p = 2$. Since P is even, $v(2z_0 - z)$ is again a wave. If q is odd, then $m = p/q$ is even, so $-1 \in \mu_m$ and $-v(2z_0 - z)$ is a wave too. In both cases $\tilde{v}(z) = (-1)^q v(2z_0 - z)$ is a wave with the same leading coefficient as v at z_0 . Reflection multiplies the normalized Laurent coefficient b_j by $(-1)^j$, so it preserves the free coefficient b_J , because $J = p + 2q$ is even. A normalized series is determined by its free coefficient, so the series of v and $\tilde{v}$ at z_0 coincide, and $\tilde{v} = v$. This is (9.1).

Place the pole at 0, so that $v(-z) = (-1)^q v(z)$. For a twisted period γ , $v(z + \gamma) = \chi(\gamma)v(z)$ and reflection give $v(z - \gamma) = \chi(\gamma)v(z)$. On the other hand $v(z - \gamma) = \chi(\gamma)^{-1}v(z)$. Hence $\chi(\gamma)^2 = 1$. $\square$

Corollary 9.2 (Limits at the two ends). *Let P be even and let v be a wave with Γ of rank one. The limits $v_{\pm}$ of v at its two ends are constant solutions of (1.1), and $v_+ = (-1)^q v_-$. Hence a front, with $v_+ \neq v_-$, requires odd q ; then n is odd, and $v_{\pm}$ are opposite nonzero constant solutions.*

Proof. By Proposition 9.1 and Lemma 5.1(c), v is a rational function of $t = e^{2\pi iz/\tau}$ that is finite at $t = 0$ and $t = \infty$; here τ generates the group of ordinary periods. So the limits exist, and all derivatives of v tend to 0 at the ends, because $D = (2\pi i/\tau) t d/dt$. Letting $\text{Re}(kz) \rightarrow \pm\infty$ in (1.1) shows that $v_{\pm}$ are constant solutions. Reflection in a pole interchanges the two ends, so taking limits in (9.1) gives $v_+ = (-1)^q v_-$. If q is odd, n is odd as in the previous proof, and a front has $v_+ = -v_- \neq v_-$, so $v_{\pm} \neq 0$. $\square$

9.2 A first-derivative term

If a first-derivative term is added to an even operator, the reflection symmetry is lost. For $n \geq 3$ and $p \geq 4$ there are then no waves at all.

Corollary 9.3 (No waves with a first-derivative term). *Let $n \geq 3$, let $p \geq 4$ be even, and let $P(D) = P_e(D) + a_1 D$, where P_e is even and $a_1 \neq 0$. Then every meromorphic solution of (1.1) is constant; the constant solutions are $v = 0$ and the roots of $v^m = -nP(0)$.*

Proof. Suppose there is a wave; then $p = mq$ with q an integer, and $K = (q)_p$. At a pole, in the normalization (3.1), induction on (3.2) gives $b_j = 0$ for odd $j < p - 1$: the even part P_e and the nonlinearity preserve the parity of Laurent indices, and the term $a_1 D$ first contributes at index $p - 1$. At that index $I(p - 1) = -nK$, because $0 \leq p - 1 - q < p$. Hence

$$b_{p-1} = -\frac{qa_1}{nK}, \quad \mathcal{R}(P) = \text{Res}_{z=0} u^m = m b_{p-1} = -\frac{pa_1}{nK} \neq 0;$$

here $\text{Res} u^m = m b_{p-1}$ because b_{p-1} is the only coefficient of odd index below p . Since $b_j = 0$ for odd $j < p - 1$,

$$v(z_0 + z) - (-1)^q v(z_0 - z) = O(z^{p-1-q}) \quad (9.2)$$

at every pole z_0 , and the exponent is positive for $n \geq 3$ and $p \geq 4$.

By Corollary 8.3(c) the wave is untwisted and Γ has rank one, and by Theorem A its poles form one orbit of Γ . Place one of them at 0. Then $u(z) - (-1)^q u(-z)$ is invariant under Γ , has no poles by (9.2), is bounded at both ends and vanishes at 0; so it is zero. Then all Laurent coefficients of odd index of u at 0 vanish, in particular $b_{p-1} = 0$, which is false. $\square$

In terms of (1.2) with zero integration constant: let $n \geq 3$, let the order $p+1$ be at least 5, and let εU_{xx} with $\varepsilon \neq 0$ be the only derivative of even order. Then no travelling wave is meromorphic and nonconstant. For example, travelling waves $U = w(x - st)$ of

$$U_t + U^2 U_x + \varepsilon U_{xx} + \delta U_{xxx} + U_{xxxx} = 0,$$

with zero integration constant, satisfy (1.1) with $n = 3$ and $P(\rho) = \rho^4 + \delta\rho^2 + \varepsilon\rho - s$, to which the corollary applies when $\varepsilon \neq 0$.

Remark 9.4. (a) In some cases the absence of waves has a local reason. For $n = 3$ and $P(D) = D^4 + a_2 D^2 + a_1 D + a_0$, the number F_J of Section 3 equals $-a_1^2 a_2 / 900$. So for $a_1 a_2 \neq 0$ no Laurent series with a pole satisfies the equation, and Proposition 4.3 alone shows that every meromorphic solution is constant, as in [25] for the Swift–Hohenberg equation without dispersion. For $a_2 = 0$ the number F_J vanishes, and the conclusion rests on Corollary 9.3.

(b) The proof of Corollary 9.3 needs the exponent $p - 1 - q$ in (9.2) to be positive; it is zero for $n = 3$, $p = 2$ and negative for $n = 2$. For $p = 2$ the conclusion does not hold: for $n = 3$, the function $v = cG$, with $G = k e^{kz} / (e^{kz} - 1)$ and $c^2 = -6$, solves $v'' - 3kv' + 2k^2 v + v^3 / 3 = 0$, and for $n = 2$ the front of the Korteweg–de Vries–Burgers equation is a wave of $v'' + a_1 v' + a_0 v + v^2 / 2 = 0$ with $a_1 \neq 0$ and $a_0 = \pm 6a_1^2 / 25$.

For $n = 2$ and even $p \geq 4$ the conclusion of Corollary 9.3 holds at least up to degree ten.

Proposition 9.5 ($n = 2$, $p \leq 10$). *Let $n = 2$, let $p \in \{4, 6, 8, 10\}$, and let $P(D) = P_e(D) + a_1 D$, where P_e is even and $a_1 \neq 0$. Then (1.1) has no waves.*

Proof. Suppose there is a wave; by Theorem A it has one of the forms (i)–(iii). Here $m = 1$ and $q = p$. As in the proof of Corollary 9.3, the recurrence gives $b_j = 0$ for odd $j < p - 1$ and $\text{Res}_{z=0} u = b_{p-1} = pa_1 / (p! - 2K) \neq 0$. So the wave is not elliptic. For a rational wave $u = \sum_{j=1}^p d_j z^{-j}$, the coefficients of z^{-1} and z^{-2} in (3.1) at infinity give $a_0 d_1 = 0$ and $a_0 d_2 - a_1 d_1 = K d_1^2$, so $d_1 \in \{0, -a_1 / K\}$. But $d_1 = b_{p-1}$, and b_{p-1} is neither of these numbers, because $(2 - p)K < 0 < p!$. For the waves that are rational in an exponential, an exact elimination in the space of Theorem A(ii), with all coefficients of P_e unknown, shows that $a_1 = 0$ (Appendix F). $\square$

We do not know a proof for all even $p \geq 4$.

9.3 The Swift–Hohenberg equation with dispersion

Theorem A turns the search for waves into a finite computation. We carry it out for the equation

$$U_t + 2U_{xx} - \sigma U_{xxx} + U_{xxxx} = \alpha U - U^3,$$

the pure cubic case of the equation studied by Kudryashov and Sinelshchikov [25]. Travelling waves $U = w(x - st)$ satisfy $P(D)w + w^3 = 0$ with

$$P(\rho) = \rho^4 - \sigma\rho^3 + 2\rho^2 - s\rho - \alpha;$$

this is (1.1) for $v = \sqrt{3}w$, so $n = 3$ and $q = 2$. The operator is not even unless $\sigma = s = 0$. We determine all waves that are rational in an exponential.

By the scaling of Section 3, a wave of P with rate 2λ comes from a wave with rate 2 of the operator $\tilde{P}$ given by

$$\tilde{P}(\rho) = \rho^4 + a_3 \rho^3 + a_2 \rho^2 + a_1 \rho + a_0, \quad P(\rho) = \lambda^4 \tilde{P}(\rho / \lambda).$$

We first find the waves with rate 2, with $a_3, \dots, a_0$ unknown, and then return to P .

By Theorem A(ii), a wave with rate 2 lies, after a translation, in one of two spaces according to its character. We place the pole at $i\pi/2$. If the multiplier under $z \mapsto z + i\pi$ is -1 , the space is spanned by $\operatorname{sech} z$ and $D \operatorname{sech} z$; if the multiplier is 1 , it is spanned by 1 , $\tanh z$ and $D \tanh z = \operatorname{sech}^2 z$. The normalized equation is $\tilde{P}(D)u = 120u^3$. We scale the amplitude so that the leading coefficient at $i\pi/2$ is $i/2$ in the first space and $-1/2$ in the second. The squares of these numbers are $\mp \frac{1}{4}$, so the coefficient 120 becomes ∓ 480 , and the wave has one of the forms

$$\begin{aligned} w_1 &= (D - \xi) \frac{1}{2} \operatorname{sech} z, & \text{with } \tilde{P}(D)w_1 &= -480 w_1^3, \\ w_2 &= \beta + \frac{1}{2} \operatorname{sech}^2 z - \frac{\xi}{2} \tanh z, & \text{with } \tilde{P}(D)w_2 &= 480 w_2^3. \end{aligned} \quad (9.3)$$

The parameter β is the midpoint of the two end values $\beta \mp \xi/2$ of w_2 as $\operatorname{Re} z \rightarrow \pm\infty$.

Both substitutions are finite computations. For w_2 , write $Y = (1 + e^{2z})^{-1}$, so that $\tanh z = 1 - 2Y$, $DY = -2Y(1 - Y)$, and w_2 is a polynomial of degree two in Y . Then $\tilde{P}(D)w_2 - 480w_2^3$ is a polynomial of degree at most six in Y . The coefficient of Y^6 vanishes by the normalization, the coefficients of $Y^5, \dots, Y^2$ determine $a_3, \dots, a_0$ in turn (Table 2, top), and what remains is affine in Y . An affine function of Y is zero exactly when it vanishes at $Y = 0$ and $Y = 1$, that is, at the two ends, where all derivatives of w_2 tend to 0. So the remaining conditions are $w_{\pm}(a_0 - 480 w_{\pm}^2) = 0$ for the two end values $w_{\pm} = \beta \mp \xi/2$. The end values are therefore equal, opposite, or one of them is zero, which puts (β, ξ) on one of the lines $\xi = 0$, $\beta = 0$, $\beta = \pm \xi/2$; on each line, inserting a_0 from the top of the table gives the equation in the lower part of Table 2. The four lines meet only at the origin $\beta = \xi = 0$, which gives the wave $\frac{1}{2} \operatorname{sech}^2 z$. For w_1 the same computation determines $a_3, \dots, a_0$ and leaves one polynomial condition on ξ .

	twisted, w_1	untwisted, w_2
a_3	-14ξ	-14ξ
a_2	$46\xi^2 + 10$	$46\xi^2 - 120\beta - 20$
a_1	$-2\xi(7\xi^2 + 10)$	$-2\xi(7\xi^2 - 300\beta - 20)$
a_0	$-14\xi^4 - 80\xi^2 - 11$	$1440\beta^2 - 120\beta\xi^2 + 480\beta - 14\xi^4 + 160\xi^2 + 64$
condition	$\xi(\xi^2 - 1)(7\xi^2 + 17) = 0$	see below

line	remaining equation for w_2 (necessary and sufficient)
$\beta = -\xi/2$	$\xi(\xi - 4)(\xi^2 - 4)(7\xi - 2) = 0$
$\beta = \xi/2$	$\xi(\xi + 4)(\xi^2 - 4)(7\xi + 2) = 0$
$\xi = 0$	$\beta(30\beta^2 + 15\beta + 2) = 0$
$\beta = 0$	$\xi(\xi^2 - 4)(7\xi^2 + 8) = 0$

Table 2: The forms (9.3) at rate $k = 2$. Top: the coefficients of $\tilde{P}$ and the remaining condition. Bottom: the remaining equation for w_2 on each of the four lines.

Altogether there are five values of ξ for the twisted form w_1 and fifteen pairs (β, ξ) for the untwisted form w_2 . To return to the physical equation, rescale by $\lambda \neq 0$: the wave is $A w_j(\lambda z)$ for the operator $P(\rho) = \lambda^4 \tilde{P}(\rho/\lambda)$, with $\Gamma = (i\pi/\lambda)\mathbb{Z}$, that is, with rate $k = 2\lambda$, and with $A^2 = 480\lambda^4$ for w_1 and $A^2 = -480\lambda^4$ for w_2 . Comparing coefficients with P gives

$$\lambda^2 a_2 = 2, \quad \sigma = -\lambda a_3, \quad s = -\lambda^3 a_1, \quad \alpha = -\lambda^4 a_0.$$

Here $a_2 \neq 0$ for all twenty parameter values, and $\sigma^2 = 2a_3^2/a_2$. Replacing z by $-z$ turns w_1 and w_2 into $-w_1$ and w_2 with $-\xi$ in place of ξ ; so the parameter values with ξ and $-\xi$, taken with the

factors λ and $-\lambda$, describe the same wave of the same equation. As s and α vary, the dispersion coefficients that admit waves rational in an exponential are

$$\begin{aligned} \text{twisted: } \sigma^2 &\in \left\{0, 7, \frac{833}{89}\right\}, \\ \text{untwisted: } \sigma^2 &\in \left\{0, \frac{392}{71}, \frac{784}{127}, \frac{1568}{239}, \frac{392}{41}, \frac{392}{11}\right\}. \end{aligned} \tag{9.4}$$

Each value of σ comes with the values of s and α determined by Table 2. By Theorem A there are no further waves rational in an exponential, and by Proposition 7.3 an equation with $\sigma \neq 0$ has at most one wave; each of the two values of σ with $\sigma^2 = 392/11$ occurs for two waves, which have different s and α .

The list agrees with [25], where the equation contains an additional quadratic term bU^2 . The untwisted waves with $\sigma \neq 0$ are its solutions with one pole per period, at $b = 0$ in conditions (25) and (26) of that paper. The twisted waves with $\sigma \neq 0$ are degenerations of its elliptic solutions with poles of two types; those solutions satisfy the condition¹ under which adding a constant to U removes the quadratic term and leaves no constant term. The stationary waves are in Section 3 of that paper.

The list also illustrates the results of the preceding sections. By Corollaries 8.2 and 8.3(b), every elliptic wave and every twisted wave rational in an exponential has $\mathcal{R}(P) = 0$, which for the Swift–Hohenberg operator reads

$$s = \frac{\sigma(392 - 39\sigma^2)}{1372};$$

this is the speed of the elliptic solutions in [25]. And the entries with $\sigma = 0$ agree with Corollary 9.3: for $\sigma = 0$ and $s \neq 0$, the operator is an even polynomial minus sD , so there are no waves; accordingly, $\sigma = -\lambda a_3 = 14\lambda\xi$ vanishes only for $\xi = 0$, and then $a_1 = 0$ and $s = 0$.

Few of these waves are real on the real axis. Suppose that a wave $w = Aw_j(\lambda z)$ of the list is. At every pole $w^2 \sim -120(z - z_0)^{-4}$, so the leading coefficient is purely imaginary. At a real pole it would be real; so w has no real pole. Its poles come in conjugate pairs, with conjugate, hence opposite, leading coefficients. An untwisted wave has the same leading coefficient at all its poles; so no untwisted wave is real. For the twisted waves with $\xi^2 = -17/7$ we have $\lambda^2 = -7/356$, so all poles lie on one line parallel to the real axis; a conjugate pair of poles would place this line on the real axis, where w has no pole; so these waves are not real either. The twisted waves with $\xi = 0$ and $\xi = \pm 1$ are real: there $\lambda^2 = 1/5$ and $1/28$, so λ and A are real, and w_1 is real on the real axis for real ξ .

10 Open questions

The bound of two waves is attained for $p = 4$ with $n = 2$ and $n = 3$ (Sections 2.6 and 8.3) and for $p = 6$ with $n = 2$ (Section 8.3). We do not know an equation with two waves for $n = 3$ and $p \geq 6$, or for $n = 2$ and $p \geq 8$.

Theorems A and B describe all waves of a single equation. The inverse question, which operators admit waves, is not treated here beyond the examples of Sections 2, 8 and 9; for $n = 2$, the operator–wave pairs with prescribed rate or lattice are counted in [30]. How to select the real bounded waves from these complex spaces, and how stable they are, remain open questions.

¹The condition is $\alpha = 2b^2/(9\kappa)$ for the nonlinear terms $\alpha U + bU^2 + \kappa U^3$; here $\kappa = -1$, and $\kappa = 30$ in [25].

A Notation

Table 3 lists the main symbols, with the place where each is introduced.

symbol	meaning	see
$D; n, m = n - 1$	d/dz ; power of the nonlinearity; phases $\omega \in \mu_m$	(1.1)
$p, q = p/m$	degree of P ; order of every pole	(1.3)
$(a)_k$	rising factorial $a(a+1) \cdots (a+k-1)$	(1.3)
wave	nonconstant solution of (1.1) that is meromorphic in $\mathbb{C}$	Section 1
$\mathcal{W}$	functions that are rational, rational in an exponential, or elliptic	Section 1
Γ, χ	group of twisted periods of a wave; its character	Definition 1.1
twisted, untwisted	with nontrivial, trivial character	Section 1
k ; ends; front	rate, $\Gamma = (2\pi i/k)\mathbb{Z}$; $\text{Re}(kz) \rightarrow \pm\infty$; wave with different limits at the ends	Section 1
a_{p-k}	coefficient of D^{p-k} in P ; it has weight k	Section 3
K, c	$K = (-1)^p(q)_p$ and $c^m = -nK$	(3.1)
u	normalized wave, $u(z) = v(z_0 + z)/c = z^{-q}(1 + O(z))$, $P(D)u = Ku^n$	(3.1)
$b_j, I(j)$	normalized Laurent coefficients; indicial polynomial	(3.2)
J, h	for even p : resonance index $J = (n+1)q$; free coefficient $h = b_J$	Lemma 3.1, (3.3)
$f(z; h), F_J$	series defined by the recurrence; $P(D)f - Kf^n = -F_J z^q$	(3.3)
ω_i, h_i	phase and free coefficient of a wave at its pole z_i	(3.4)
$S, d(z)$	pole set of a solution; distance to it	Section 4
$g_r, G = g_m$	functions with simple poles on $\Gamma = (2\pi i/k)\mathbb{Z}$	(1.4)
V_j^χ, g_χ	functions of the second kind on a lattice, poles of order $\leq j$; element of V_1^χ with residue 1	Section 5.2
$\wp(z; g_2, g_3), \wp(z; \Gamma)$	Weierstrass function with $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$; that of the lattice Γ	Section 2.6
$\hat{\alpha}$	principal-part polynomial of a Laurent series α	Section 6
$E, F; \varphi, \Phi$	two expressions in u ; the combination (6.1) on $f(z; h)$ and on a wave	(6.1)
H, H_n	$H(u) = uu'' - \frac{q+1}{q}(u')^2$ and $H_n = u^{n-2}H$	(6.2), Section 6.4
Δ_1	slope in h of the coefficient $[z^{p-2q-1}]\varphi$, for $n \geq 4$	Proposition 6.4
$[h^k], [z^j]$	coefficient of h^k , of z^j in a series	Section 6
$\psi, \kappa, \Delta_2; X$	leading coefficients of the terms in h^2 , for $n = 2, 3$; $\psi = \binom{n}{2}/(X - n)$	(6.4)–(6.6)
a_*	leading coefficient of $\hat{E}$ for $E = u$	Proposition 6.6
A_i, B_i	$[z^i]\varphi = A_i h + B_i$ for the pair (u, H_n) and $i \leq J - 2$	(6.7)
$T; Q_P$	entire function whose values at the poles do not contain h ; quadratic polynomial with $Q_P(h) = 0$	Proposition 6.6
c_0, c_1, c_{j-1}	numerical factors of A_0, A_1, A_{j-1}	Lemma 7.2
$\ell_j, \mathcal{R}(P)$	coefficient of z^{j-p} in u^m ; the residue $\mathcal{R}(P) = \ell_{p-1}$	(8.1), Corollary 8.2
Λ_r	invariant function with $\Lambda_r = z^{-r} + O(z)$	Section 8

Table 3: Notation used throughout.

B Elliptic functions of the second kind

We prove Lemma 5.3. Recall that V_j^χ is the space of meromorphic functions w with $w(z + \gamma) = \chi(\gamma)w(z)$ for γ in the lattice Γ , whose poles lie in Γ and have order at most j .

Proof of Lemma 5.3. A function w in V_0^χ is entire, and $|w|$ is periodic, so w is bounded and hence constant; a nonzero constant has trivial character. Thus $V_0^\chi = 0$ if χ is not trivial.

For existence, let σ be the Weierstrass sigma function of Γ , and let $\eta: \Gamma \rightarrow \mathbb{C}$ be the quasi-period homomorphism of the zeta function, $\zeta(z + \gamma) = \zeta(z) + \eta(\gamma)$. The function $e^{az}\sigma(z - z_\chi)/\sigma(z)$ has the multiplier $\exp(a\gamma - \eta(\gamma)z_\chi)$ under $z \mapsto z + \gamma$. If γ_1, γ_2 is a basis of Γ , Legendre's relation $\eta(\gamma_1)\gamma_2 - \eta(\gamma_2)\gamma_1 = \pm 2\pi i$ shows that $(a, z_\chi) \mapsto (a\gamma_1 - \eta(\gamma_1)z_\chi, a\gamma_2 - \eta(\gamma_2)z_\chi)$ is a linear bijection of $\mathbb{C}^2$; so a and z_χ can be chosen to realize any character. If χ is not trivial, then $z_\chi \notin \Gamma$: otherwise the function would be a constant multiple of some e^{bz} , whose multipliers $e^{b\gamma}$ have finite order for all γ in a lattice only if $b = 0$, and then χ would be trivial. So the function has a simple pole at 0, and dividing by its residue gives g_χ . The derivatives $D^i g_\chi$ have leading terms $(-1)^i i! z^{-i-1}$, so a suitable combination of them has any prescribed principal part at 0. The difference from a given element of V_j^χ lies in $V_0^\chi = 0$. This proves the basis property and the uniqueness.

For the trivial character, the residue at the only pole of an elliptic function is zero, and the derivatives of $\wp$ remove all higher principal coefficients, leaving a constant. $\square$

For $m = 2$ the functions g_χ of the three characters of order two are

$$g_{\chi_i} = \sqrt{\wp - e_i} \quad (i = 1, 2, 3), \quad (\text{B.1})$$

where e_1, e_2, e_3 are the values of $\wp$ at the three half-periods of Γ and the root is chosen with leading term z^{-1} . Up to scale these are the Jacobi functions $\text{cs}, \text{ds}, \text{ns}$; translation by the half-period that is called iK' in Jacobi's notation turns them into multiples of $\text{dn}, \text{cn}, \text{sn}$ [32, §§22.4, 23.6].

C The terms linear in the free coefficient

This appendix computes the constants of Lemma 7.2. Its formulas, in particular (C.4) and (C.5), are also checked by exact computation in the script `check_linear_terms.py` (Appendix F). We keep the assumptions of the lemma: $p \geq 4$ is even, $q \geq 2$, $E = u$ and $F = H_n$. Fix an operator P and a value of h , and write

$$f = f(z; h), \quad f_h = \frac{\partial f}{\partial h}, \quad \eta = H_n(f), \quad \eta_h = \frac{\partial \eta}{\partial h};$$

for series in z , a prime denotes the derivative with respect to z . By Lemma 3.2 and the table (6.8), $f_h = z^{nq}(1 + O(z))$ and $\eta_h = J(J+1)z^{q-2}(1 + O(z))$ are power series. The principal parts of f and η do not depend on h , so

$$\frac{\partial \varphi}{\partial h} = \widehat{E}(D) \eta_h - \widehat{F}(D) f_h, \quad (\text{C.1})$$

where $\widehat{E}$ and $\widehat{F}$ are the principal-part polynomials of f and η . By (6.7), A_i is the coefficient of z^i in $\partial \varphi / \partial h$ for $i \leq J - 2$, at every value of h .

C.1 Residue formulas

Let α be a Laurent series with finite principal part $\sum_r \alpha_{-r} z^{-r}$, and $g = \sum_k g_k z^k$ a power series. By the definition of $\hat{\alpha}$,

$$\begin{aligned} [z^0] \hat{\alpha}(D)g &= \sum_r (-1)^{r-1} \alpha_{-r} g_{r-1} = \text{Res}_{z=0} \alpha(z) g(-z), \\ [z^1] \hat{\alpha}(D)g &= \sum_r (-1)^{r-1} r \alpha_{-r} g_r = \text{Res}_{z=0} \alpha'(z) g(-z). \end{aligned}$$

With (C.1) this gives, for every value of h ,

$$A_0 = \text{Res}_{z=0} [f(z) \eta_h(-z) - \eta(z) f_h(-z)], \quad A_1 = \text{Res}_{z=0} [f'(z) \eta_h(-z) - \eta'(z) f_h(-z)]. \quad (\text{C.2})$$

C.2 Even operators

We first show that A_1 contains no term a_{p-2} .

Lemma C.1. *If P is even, then $A_1 = 0$.*

Proof. For even P the recurrence preserves the parity of Laurent indices, so $b_j(h) = 0$ for odd j . Hence $f_h(-z) = (-1)^{nq} f_h(z)$ and $\eta_h(-z) = (-1)^q \eta_h(z)$, and $(-1)^{nq} = (-1)^q$ because p is even. So (C.2) gives

$$A_1 = (-1)^q \text{Res}_{z=0} [f' \eta_h - \eta' f_h].$$

Let $\mathcal{L}$ be the linearization of H_n at f :

$$\mathcal{L}g = f^{n-1} g'' - \frac{2(q+1)}{q} f^{n-2} f' g' + \theta g$$

with a Laurent series θ . Then $\eta_h = \mathcal{L}f_h$, and $\eta' = \mathcal{L}f'$ because H_n has constant coefficients. With the Wronskian $W = f' f'_h - f'' f_h$ we get

$$f' \eta_h - \eta' f_h = f^{n-1} W' - \frac{2(q+1)}{q} f^{n-2} f' W.$$

The residue of a derivative vanishes. Integrating by parts twice, and using $(n-1) + 2(q+1)/q = (J+2)/q$, we find

$$\text{Res}[f' \eta_h - \eta' f_h] = -\frac{J+2}{q} \text{Res}[f^{n-2} f' W] = \frac{J+2}{q} \text{Res}[f_h \Theta],$$

where

$$\Theta = (n-2) f^{n-3} (f')^3 + 3 f^{n-2} f' f'';$$

the last step holds because $f^{n-2} f' W = (f^{n-2} (f')^2 f_h)' - f_h \Theta$.

Now $(f^n)''' = n f^{n-1} f''' + n(n-1) \Theta$. Put $\mathcal{M} = P(D) - nK f^{n-1}$, and recall from Section 3 that $P(D)f - K f^n = -F_J z^q$, where the number F_J does not depend on h . Differentiating this identity with respect to h gives $\mathcal{M}f_h = 0$, and differentiating it three times with respect to z gives

$$\mathcal{M}f_h''' = n(n-1)K \Theta - F_J (z^q)'''.$$

The product of f_h with a polynomial has no residue, so $n(n-1)K \text{Res}[f_h \Theta] = \text{Res}[f_h \mathcal{M}f_h''']$. Finally, for an even operator P , integration by parts gives $\text{Res}[\alpha P(D)\beta] = \text{Res}[\beta P(D)\alpha]$ for all Laurent series α, β . Hence $\text{Res}[f_h \mathcal{M}f_h'''] = \text{Res}[f_h''' \mathcal{M}f_h] = 0$, and $A_1 = 0$. $\square$

By Lemma 7.2(a), $A_1 = c_1 a_{p-1}^2 + c' a_{p-2}$ with numbers c_1, c' . For the even operator $D^p + a_{p-2} D^{p-2}$, Lemma C.1 gives $c' a_{p-2} = 0$, so $c' = 0$.

C.3 An explicit family

We now compute c_0 and c_1 . For $\varepsilon \neq 0$ consider the operator

$$P_\varepsilon(\rho) = \prod_{i=0}^{p-1} (\rho + (q+i)\varepsilon), \quad a_{p-1} = \varepsilon \sum_{i=0}^{p-1} (q+i) = \frac{1}{2} \varepsilon p (J-1).$$

Put $\zeta = e^{-\varepsilon z}$, so that $D = -\varepsilon \zeta d/d\zeta$. For a function g of ζ we have $(D + (q+i)\varepsilon)[\zeta^{q+i}g] = -\varepsilon \zeta^{q+i+1} dg/d\zeta$. Applying the factors with $i = 0, \dots, p-1$ in turn gives

$$P_\varepsilon(D)[\zeta^q g(\zeta)] = \varepsilon^p \zeta^{q+p} \frac{d^p g}{d\zeta^p}.$$

Hence, if $Y(x)$ satisfies $Y^{(p)} = KY^n$, then $u = (-\varepsilon)^q \zeta^q Y(\zeta - 1)$ satisfies $P_\varepsilon(D)u = Ku^n$, because $q+p = nq$ and p is even. The operator D^p admits the wave x^{-q} , so its series (3.3) is a formal solution of $Y^{(p)} = KY^n$ for every value $\tilde{h}$ of the free coefficient, and by Lemma 3.2 it equals $x^{-q}(1 + \tilde{h}x^J + O(\tilde{h}^2))$. Therefore, with $x = e^{-\varepsilon z} - 1$,

$$(-\varepsilon)^q (1+x)^q x^{-q} (1 + \tilde{h}x^J + O(\tilde{h}^2)) \quad (\text{C.3})$$

is a formal solution of $P_\varepsilon(D)u = Ku^n$. It starts with z^{-q} , so it is the series $f(z; h)$ of P_ε , where h is its coefficient of z^{J-q} . The term linear in $\tilde{h}$ is $(-\varepsilon)^q (1+x)^q x^{nq} \tilde{h} = \varepsilon^J z^{nq} (1 + O(z)) \tilde{h}$, because $x = -\varepsilon z(1 + O(z))$ and $(-\varepsilon)^{q+nq} = \varepsilon^J$, and the terms of higher degree in $\tilde{h}$ are $O(z^{2J-q})$. So $h = h_0 + \varepsilon^J \tilde{h}$ for a constant h_0 . For $\tilde{h} = 0$ it is the wave $u = (\varepsilon/(e^{\varepsilon z} - 1))^q$ of P_ε .

We evaluate (C.2) at $h = h_0$, that is, at $\tilde{h} = 0$, where $\partial/\partial h = \varepsilon^{-J} \partial/\partial \tilde{h}$. By (6.3), $H_n(f) = -q y^{-nq-1} y''$ with $y = f^{-1/q}$, and (C.3) gives $y = -x(1 + \tilde{h}x^J)^{-1/q}/(\varepsilon(1+x)) + O(\tilde{h}^2)$. Using $D = -\varepsilon(1+x) d/dx$, a direct computation yields, at $\tilde{h} = 0$,

$$\begin{aligned} f &= (-\varepsilon)^q (1+x)^q x^{-q}, & \varepsilon^J f_h &= (-\varepsilon)^q (1+x)^q x^{nq}, & \eta &= q \varepsilon^2 (-\varepsilon)^{nq} (1+x)^{nq} x^{-nq-1}, \\ \varepsilon^J \eta_h &= \varepsilon^2 (-\varepsilon)^{nq} (1+x)^{nq} x^{q-2} (J(J+1) + (2J^2 + J + nq)x + J^2 x^2). \end{aligned} \quad (\text{C.4})$$

We compute the residues in the variable x . Here $dz = -dx/(\varepsilon(1+x))$, and replacing z by $-z$ replaces x by $-x/(1+x)$, so that a product $(1+x)^\mu x^\nu$ becomes $(-1)^\nu (1+x)^{-\mu-\nu} x^\nu$. Thus

$$\varepsilon^J f_h(-z) = (-1)^q (-\varepsilon)^q \frac{x^{nq}}{(1+x)^J}, \quad \varepsilon^J \eta_h(-z) = (-1)^q \varepsilon^2 (-\varepsilon)^{nq} \frac{x^{q-2} (J(J+1) + qx - nqx^2)}{(1+x)^J},$$

and $f' = q \varepsilon (-\varepsilon)^q (1+x)^q x^{-q-1}$, $\eta' = q \varepsilon^3 (-\varepsilon)^{nq} (1+x)^{nq} x^{-nq-2} (nq+1+x)$. Since $(-\varepsilon)^{q+nq} = \varepsilon^J$, this gives

$$\begin{aligned} A_0 &= (-1)^{q+1} \varepsilon \operatorname{Res}_{x=0} \left[\frac{J(J+1) + qx - nqx^2}{x^2(1+x)^{nq+1}} - \frac{q}{x(1+x)^{q+1}} \right] \\ &= (-1)^q \varepsilon J(J+1)(nq+1), \\ A_1 &= (-1)^{q+1} q \varepsilon^2 \operatorname{Res}_{x=0} \left[\frac{J(J+1) + qx - nqx^2}{x^3(1+x)^{nq+1}} - \frac{nq+1+x}{x^2(1+x)^{q+1}} \right] \\ &= (-1)^{q+1} \varepsilon^2 \frac{1}{2} q J(J+1)(nq+1)(nq+2). \end{aligned}$$

Both are nonzero. Dividing by a_{p-1} and by a_{p-1}^2 , and using $J/p = (n+1)/(n-1)$, we obtain the constants c_0 and c_1 of Lemma 7.2(b).

C.4 The first odd coefficient

It remains to compute c_{j-1} and to show that it is not zero. Let j be odd with $3 \leq j \leq p-1$, and let $P = D^p + a_{p-j}D^{p-j}$. The number c_{j-1} is the coefficient of a_{p-j} in A_{j-1} , so we compute modulo a_{p-j}^2 ; we write $a = a_{p-j}$ for short. By Lemma 3.2, modulo a^2 and h^2 ,

$$f(z; h) = z^{-q}(1 + \beta_j a z^j + h z^J + \gamma_j a h z^{J+j})$$

with numbers β_j and γ_j . The recurrence (3.2) at the indices j and $J+j$ reads

$$I(j) \beta_j = (q)_{p-j}, \quad I(J+j) \gamma_j = -\frac{(nq)!}{(q+j)!} + (n-1) \frac{(nq)!}{q!} \beta_j.$$

By the proof of Lemma 3.1, $I(j) < 0$ and $I(J+j) > 0$. So $\beta_j < 0$ and $\gamma_j < 0$.

We use (C.1) at $h = 0$. Let $\mathcal{B}$ be the bilinear form of H , as in the proof of Proposition 6.6. A direct computation gives $2\mathcal{B}(z^{i-q}, z^{nq}) = ((J-i)(J-i+1) - 2ni)z^{p-2+i}$, and Lemma 6.3 gives $2\mathcal{B}(z^{-q}, z^{nq+i}) = (J+i)(J+i+1)z^{p-2+i}$ and $H(z^{-q}(1 + \beta_j a z^j)) = j(j+1)\beta_j a z^{j-2q-2}$. Hence, at $h = 0$ and modulo a^2 ,

$$\begin{aligned} \eta &= j(j+1)\beta_j a z^{j-nq-2}, & \eta_h &= J(J+1)z^{q-2} + (s_j \beta_j + (J+j)(J+j+1)\gamma_j) a z^{q-2+j}, \\ s_j &= (n-2)(J(J+1) + j(j+1)) + (J-j)(J-j+1) - 2nj; \end{aligned}$$

the three parts of s_j come from f^{n-2} , from the factor $f^{n-3}f_h H(f)$, and from $\mathcal{B}(f, f_h)$. The principal-part polynomials are, modulo a^2 ,

$$\widehat{E}(\rho) = \frac{(-\rho)^{q-1}}{(q-1)!} + \beta_j a \frac{(-\rho)^{q-j-1}}{(q-j-1)!}, \quad \widehat{F}(\rho) = j(j+1)\beta_j a \frac{(-\rho)^{nq+1-j}}{(nq+1-j)!},$$

where the second term of $\widehat{E}$ is absent if $j \geq q$. Reading off the coefficient of $a z^{j-1}$ in (C.1), with j odd and p even, we obtain

$$\begin{aligned} (-1)^q c_{j-1} &= - \binom{q+j-2}{j-1} (s_j \beta_j + (J+j)(J+j+1)\gamma_j) \\ &\quad + J(J+1) \binom{q-2}{j-1} \beta_j - j(j+1) \binom{nq}{j-1} \beta_j, \end{aligned} \tag{C.5}$$

where $\binom{q-2}{j-1} = 0$ if $j \geq q$. Up to this point $n \geq 2$ is arbitrary.

Since $\beta_j < 0$ and $\gamma_j < 0$, we can write (C.5) as

$$\begin{aligned} (-1)^q c_{j-1} &= |\beta_j| \left[\binom{q+j-2}{j-1} s_j - J(J+1) \binom{q-2}{j-1} \right] \\ &\quad + \binom{q+j-2}{j-1} (J+j)(J+j+1) |\gamma_j| + j(j+1) \binom{nq}{j-1} |\beta_j|. \end{aligned}$$

The last two terms are positive. So $(-1)^q c_{j-1} > 0$ will follow from

$$\binom{q+j-2}{j-1} s_j \geq J(J+1) \binom{q-2}{j-1}, \tag{C.6}$$

which we now prove for $n = 3$ and for $n = 2$. For $n = 3$ we have $J = 4q$, $j \leq 2q - 1$ and

$$s_j = J(J+1) + j(j-5) + (J-j)(J-j+1) > J(J+1),$$

because $j(j-5) \geq -6$ and $J-j \geq 2q+1 \geq 5$. Since $\binom{q+j-2}{j-1} \geq \binom{q-2}{j-1}$, this proves (C.6). For $n = 2$ we have $J = 3q$, $3 \leq j \leq q-1$ and

$$s_j = J(J+1) - j(6q+5-j) \geq J(J+1) \left(1 - \frac{2j}{3q}\right) > 0,$$

because $j(6q+5-j) \leq j(6q+2)$ and $J(J+1) = 3q(3q+1)$. Moreover

$$\binom{q+j-2}{j-1} / \binom{q-2}{j-1} = \prod_{i=0}^{j-2} \left(1 + \frac{j}{q-2-i}\right) \geq 1 + \frac{j(j-1)}{q},$$

since each of the $j-1$ factors is at least $1 + j/q$. The product of the two lower bounds is

$$\left(1 + \frac{j(j-1)}{q}\right) \left(1 - \frac{2j}{3q}\right) = 1 + \frac{j}{q} \left(j-1 - \frac{2}{3} - \frac{2j(j-1)}{3q}\right) \geq 1 + \frac{j}{q} \cdot \frac{j-3}{3} \geq 1,$$

where we used $j < q$ in the first inequality. This proves (C.6) for $n = 2$. In both cases $c_{j-1} \neq 0$, which completes the proof of Lemma 7.2.

D Degrees one and two

Theorems A and B assume $p \geq 3$. The one-pole conclusion extends to the following cases of low degree.

Proposition D.1. *Every wave has exactly one pole modulo Γ , and hence one of the forms of Proposition 5.4, in each of the following cases: (a) $p = 1$; (b) $P(D) = D^2 + a_0$; (c) $P(D) = D^2 + a_1D + a_0$ and the wave is elliptic. In case (c) necessarily $a_1 = 0$.*

Proof. (a) Since p is odd, the normalized series is unique; apply Corollary 5.2.

(b) Pole balance leaves $(n, q) = (2, 2)$ or $(3, 1)$, and $J = 2q + 2$. The energy

$$\mathcal{E} = \frac{(v')^2}{2} + \frac{a_0 v^2}{2} + \frac{v^{n+1}}{n(n+1)}$$

is constant along every solution. At a pole write $v = cz^{-q}(1 + \sum_{j \geq 1} b_j z^j)$ with $h = b_J$. The constant $\mathcal{E}$ equals the constant term of its Laurent expansion at the pole, which gives

$$\frac{\mathcal{E}}{c^2} = -q(2q+3)h + \mathcal{E}_0,$$

where $\mathcal{E}_0$ depends only on $b_1, \dots, b_{J-1}$, hence only on P . Indeed, the coefficient of h is $-q(q+2)$ from $(v')^2/2$ plus $-q(q+1)$ from the last term, while the h -term of $a_0 v^2/2$ has a positive power of z . In both admissible cases c^2 is the same for all leading coefficients. Hence h is the same at all poles, and Corollary 5.2 applies.

(c) Pole balance again leaves $(n, q) = (2, 2)$ or $(3, 1)$. At each pole, the recurrence gives $c^m = -nq(q+1)$ and $b_1 = -a_1/(n+3)$. The elliptic function v^m has a double pole at each pole of v , with residue

$$\text{Res } v^m = mc^m b_1 = \frac{mnq(q+1)}{n+3} a_1,$$

the same at every pole. The residues in a period cell add up to zero, so $a_1 = 0$, and (b) applies. $\square$

Part (b) is the case $p = 2$ of Lemma E.1 below, since $I'(J) = 2q + 3$ when $p = 2$. In two respects degree two differs from higher degree.

Infinitely many waves. A single equation can have infinitely many waves, so the bound of Theorem B requires $p \geq 3$. For $n = 2$ and $P(D) = D^2 + a_0$, the functions

$$v = -12 \wp(z; \frac{a_0^2}{12}, g_3) - a_0, \quad g_3 \in \mathbb{C}, \quad 27g_3^2 \neq (\frac{a_0^2}{12})^3,$$

solve $v'' + a_0v + v^2/2 = 0$, by the identity $\wp'' = 6\wp^2 - g_2/2$. Their free coefficient is $h = g_3/28$, so different values of g_3 give different normalized series, and the corresponding waves are not related by translation or phase change. They still have one pole modulo Γ , as Proposition D.1(b) asserts.

Waves outside $\mathcal{W}$. Equations with $a_1 \neq 0$ can have meromorphic solutions outside $\mathcal{W}$ [6, Lemmas 4.4 and 4.5]. For $k \neq 0$ and all constants ζ_0 and g_3 , the function $v = -12k^2e^{-2kz}\wp(e^{-kz} - \zeta_0; 0, g_3)$ is a meromorphic solution of $v'' + 5kv' + 6k^2v + v^2/2 = 0$ [1], [6, Lemma 4.4]. For $g_3 \neq 0$ it lies outside $\mathcal{W}$, because it has too many poles: they are the points with $e^{-kz} \in \zeta_0 + \Lambda$, where Λ is the period lattice of $\wp(\cdot; 0, g_3)$, and their number in $|z| \leq r$ grows exponentially with r , whereas a function of $\mathcal{W}$ has $O(r^2)$ poles in $|z| \leq r$. The expression $(v' + 2kv)^2 + v^3/3 = -144g_3k^6e^{-6kz}$ has no poles, and for $g_3 \neq 0$ it is not constant. So the hypothesis on the values at the poles in Lemma 4.4 cannot be dropped.

E Polynomial nonlinearities

Moving a constant solution of a general polynomial equation to 0 and absorbing the linear term into P gives

$$P(D)v + B(v) = 0, \quad B(v) = \frac{v^n}{n} + B_*(v), \quad (\text{E.1})$$

where B_* is a polynomial that contains only the powers $v^2, \dots, v^{n-1}$. The leading balance at a pole is unchanged: $p = mq$ and $c^m = -nK$. The Laurent recurrence has the same indicial polynomial I ; the lower terms of B change only the right-hand sides, which now also depend on the choice of c . The phase changes that preserve (E.1) form the group $\mu_{m'}$, where m' is the largest divisor of m such that B contains only powers v^j with $j \equiv 1 \pmod{m'}$. For the pure power, $m' = m$.

For an even operator $P(D) = D^p + \sum_{0 \leq i < p/2} a_{2i}D^{2i}$, put

$$\mathcal{Q}_i(v) = \sum_{a=0}^{i-2} (-1)^a v^{(a+1)} v^{(2i-1-a)} + \frac{(-1)^{i-1}}{2} (v^{(i)})^2,$$

so that $D\mathcal{Q}_i(v) = v'v^{(2i)}$; for $i = 1$ the sum is empty. The energy

$$\mathcal{E}(v) = \mathcal{Q}_{p/2}(v) + \sum_{1 \leq i < p/2} a_{2i} \mathcal{Q}_i(v) + \frac{a_0}{2} v^2 + \int_0^v B(x) dx \quad (\text{E.2})$$

satisfies $D\mathcal{E}(v) = v'(P(D)v + B(v))$, so it is constant along every solution. This first integral fixes the free coefficient, as in Eremenko, Liao, and Ng [15, Lemma 4]. Demina [8] classifies the meromorphic integrals of autonomous equations with two dominant monomials.

Lemma E.1 (The energy fixes the free coefficient). *Let P be even, and fix a root c of $c^m = -nK$. The Laurent recurrence with leading coefficient c is compatible at $J = p + 2q$, and its formal solutions form a one-parameter family $v = cz^{-q}(1 + \sum_{j \geq 1} b_j z^j)$ with free coefficient $h = b_J$. Along this family*

$$\mathcal{E}(v) = -qc^2 I'(J) h + \mathcal{E}_0, \quad I'(J) = (q+1)_p \sum_{i=1}^p \frac{1}{q+i} > 0, \quad (\text{E.3})$$

where $\mathcal{E}_0$ depends only on P , B and c . Hence a leading coefficient and a value of the energy determine the Laurent series.

Proof. Construct the series up to index $J - 1$. For this truncated series the lowest term of $P(D)v + B(v)$ that can be nonzero is Ωz^q , and the residue of $v'(P(D)v + B(v))$ is then $-qc\Omega$. This is the residue of the derivative $D\mathcal{E}$, so it vanishes, and $\Omega = 0$. The coefficient b_J is therefore free, and all later coefficients are determined.

Every term of $\mathcal{E}$ has a pole of order at most J . A factor h comes with the power z^J . So h enters the constant term only through terms of pole order J in which all other factors are at leading order, and it enters linearly. To compute the coefficient of h , perturb $v = cz^{-q}(1 + \varepsilon z^j)$. The top linear term of the equation is $\varepsilon c I(j) z^{j-q-p}$; write the corresponding term of $\mathcal{E}$ as $\varepsilon L(j) z^{j-J}$. The identity $D\mathcal{E} = v'(P(D)v + B(v))$ gives $(j - J)L(j) = -qc^2 I(j)$ as polynomials in j , so $L(J) = -qc^2 I'(J)$. The formula for $I'(J)$ follows by differentiating the product in $I(j)$, whose factors at $j = J$ are $q + p, \dots, q + 1$. Finally, since $D\mathcal{E} = 0$ formally, the constant Laurent coefficient of $\mathcal{E}$ is its value. $\square$

The theorem below needs Lemma 5.1 for the solutions of (E.1) and for subgroups of μ_m . The proof of that lemma uses only Lemma 4.2 and the finiteness of μ_m . So the lemma holds for every nonconstant meromorphic solution v of (4.1) and every finite group $U \subset \mathbb{C}^*$, with μ_m replaced by U and Γ by the group

$$\Gamma_U = \{\gamma \in \mathbb{C} : v(z + \gamma) = \omega v(z) \text{ for some } \omega \in U\};$$

since $v \not\equiv 0$, the multiplier $\omega = \chi(\gamma)$ is unique, and $\chi: \Gamma_U \rightarrow U$ is a homomorphism.

Theorem E.2 (Polynomial nonlinearity; cf. [5, 14, 39]). *Let B have the symmetry group $\mu_{m'}$, and assume that p is odd or P is even. Then every nonconstant meromorphic solution of (E.1) belongs to $\mathcal{W}$. Put $\mathcal{H} = \mu_{m'}$ if p is odd, and $\mathcal{H} = \mu_{m'} \cap \{\pm 1\}$ if P is even. Two poles whose leading coefficients have ratio $\omega \in \mathcal{H}$ differ by a twisted period with multiplier ω , and the poles form at most $m/|\mathcal{H}|$ cosets of $\Gamma_{\mathcal{H}}$.*

Proof. For odd p the recurrence determines the Laurent series from its leading coefficient. For even P the energy of a solution is constant, so by Lemma E.1 each leading coefficient carries at most one Laurent series among the poles of the solution.

Let z_1, z_2 be poles with $c_2 = \omega c_1$, $\omega \in \mathcal{H}$, and compare $v(z_1 + z)$ with $\omega^{-1}v(z_2 + z)$. The second function is again a solution, because ω is a symmetry, and the two have the same leading coefficient. For odd p this identifies the two Laurent series. For even P , the symmetry gives $\mathcal{E}(\omega v) = \omega^2 \mathcal{E}(v)$, since every quadratic term scales by ω^2 and $\int_0^{\omega v} B = \omega^2 \int_0^v B$; for $\omega = \pm 1$ the two energies agree, and the energy lemma identifies the series. By the identity theorem, $z_2 - z_1$ is a twisted period with multiplier ω .

The m leading coefficients form $m/|\mathcal{H}|$ orbits under $\mathcal{H}$, and two poles whose leading coefficients lie in the same orbit differ by an element of $\Gamma_{\mathcal{H}}$, as just shown. Lemma 5.1, in the form stated

before the theorem with $U = \mathcal{H}$, gives membership in $\mathcal{W}$ and the bound on the number of cosets. $\square$

For the pure power and an even operator with $n \leq 3$ we have $\mu_m \subset \{\pm 1\}$; so the theorem gives one pole orbit, as Lemma 7.1 does.

Remark E.3. For an even operator and a polynomial nonlinearity, the reflection argument of Proposition 9.1 applies whenever $(-1)^q \in \mu_m$: the reflected series has the same leading coefficient and the same free coefficient, because J is even. So reflection symmetry holds for every even q , and for odd q when B is odd. This alone does not relate different orbits of leading coefficients.

F Computer checks

The theorems are proved in the text for all n and q . Exact computations with Python 3 and SymPy check the displayed identities, the constants of Sections 6 and 7 and of Appendix C, and the examples. The numbers quoted in the examples without derivation, such as the entries of Table 2, the number F_J in Remark 9.4(a) and the coefficient A_5 in Section 7.3, are obtained by finite computations that can be done by hand. The only statement of the paper that rests on a computer elimination is Proposition 9.5; the script `check_drift_n2.py` carries out the elimination for $p = 4, 6, 8, 10$ and shows that the statement does not hold for $p = 2$. The scripts are the ancillary files (`anc/`); the command `python3 run_all.py` in that folder runs all of them, and `anc/README.md` describes each script.

Use of AI tools

This paper was prepared with substantial use of generative AI, disclosed as recommended by the Leiden Declaration on Artificial Intelligence and Mathematics.² The main mathematical ideas were produced by `gpt-6-astra` (OpenAI) in dialogue with the author; parts of the proofs, of the verification and of the preparation of the text were carried out with Claude Fable 5.1 and Claude Opus 5.5 (Anthropic). No automated system is an author; responsibility for the results and for the attribution of prior work rests with the author.

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²<https://leidendeclaration.ai/>, 2 June 2026.

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